OP Shutdown lectures:
RF Beam Control

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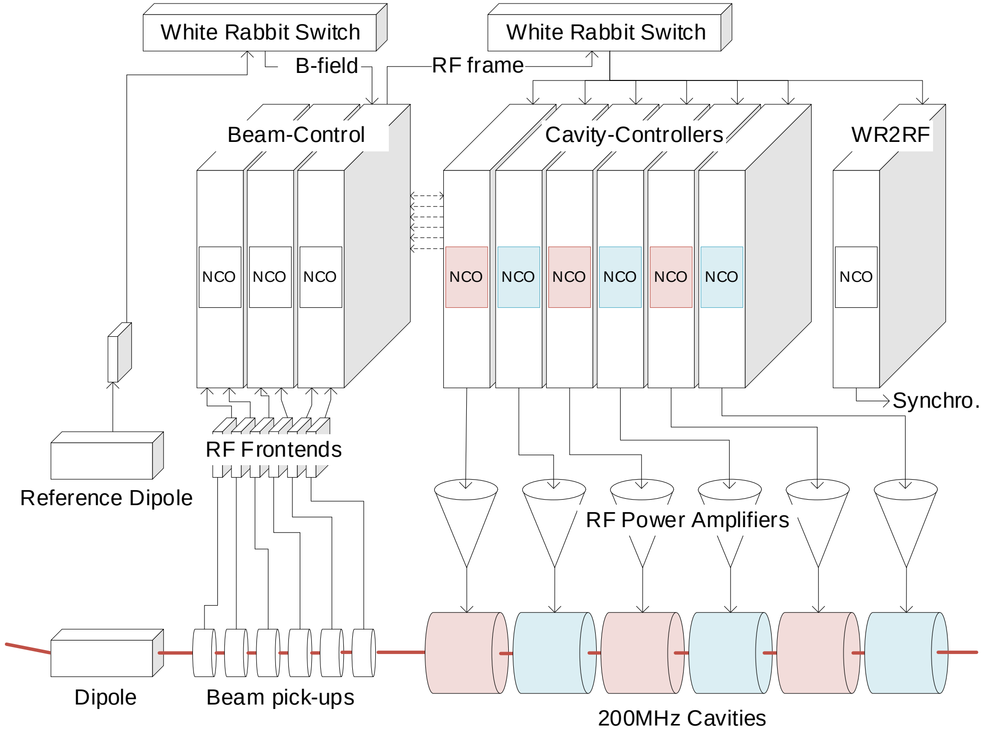
Overview

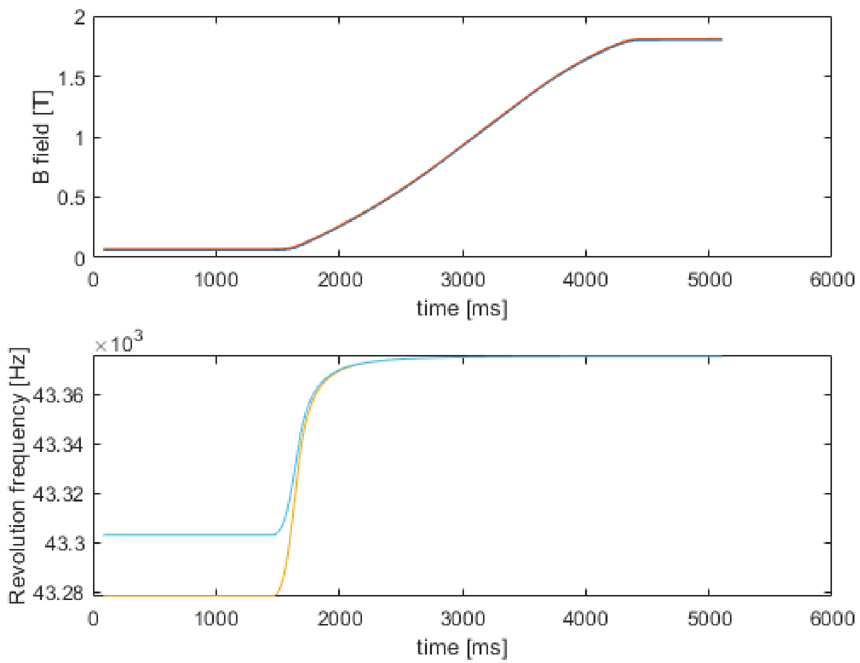
- **Go through the operational tools to setup and debug the RF beam control**
  - Revolution frequency
  - Rephasing
  - Beam loops and RF pick-ups
  - RF manipulations for ions
  - RF gymnastics
- **Overview of the Low Level RF hardware and concepts**

Revolution Frequency Generation

Revolution frequency generation

- **The revolution frequency is computed by the Beam Control (LLRF) from a measurement or a function of the bending magnetic field**
- **It also requires SPS machine parameters such as the radius, transition energy, particle mass over charge, radial steering**





$$\gamma^2 = 1 + \frac{B^2}{\left(\frac{m}{q}\right)^2 \left(\frac{E_0}{\rho c e}\right)^2}$$

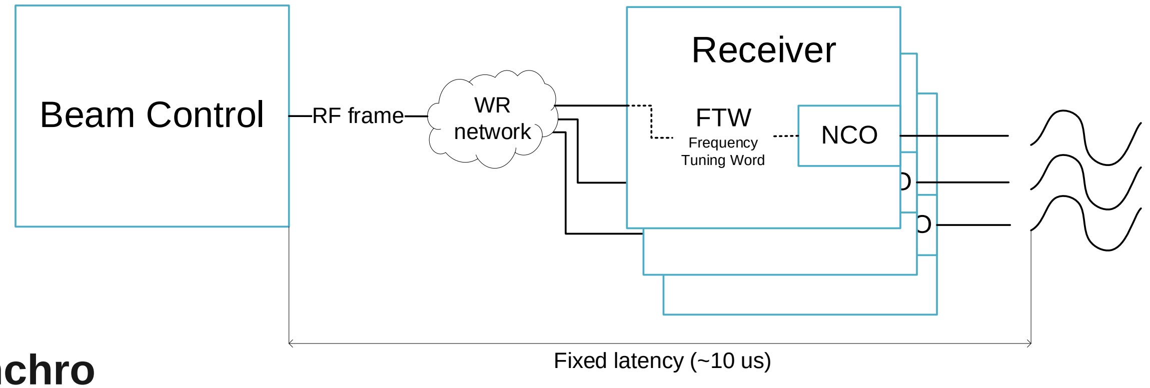
$$f_{rev} = f_{rev,inf}\sqrt{1 - \frac{1}{\gamma^2}}$$

What is B0?

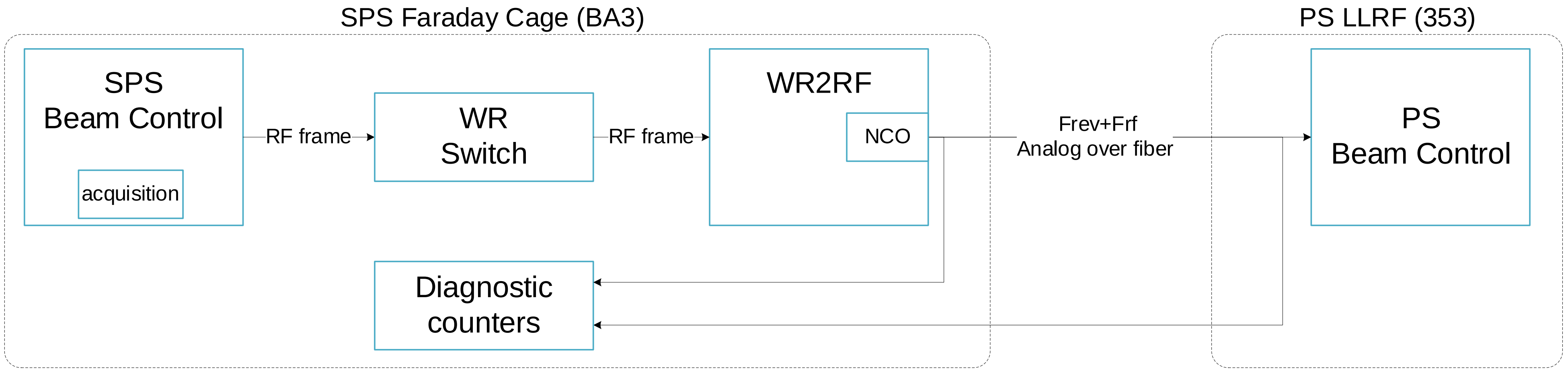
- It replaces the value of the measurement at flat bottom to provide a defined and stable frequency for injection
- Until BtrainEnable timing

Quick reminder on the NCO

- **Numerically Controlled Oscillator**
- **Synchronized over White Rabbit (WR) network (Ethernet + precision timing)**

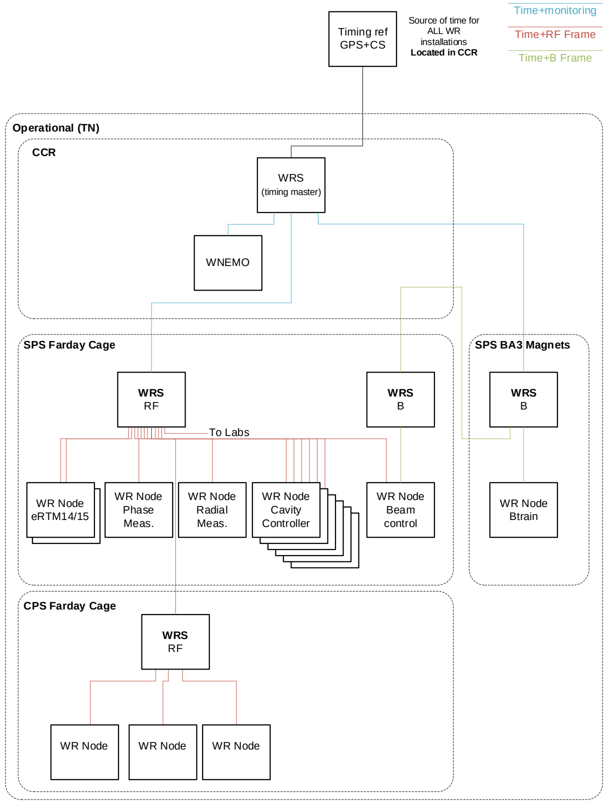


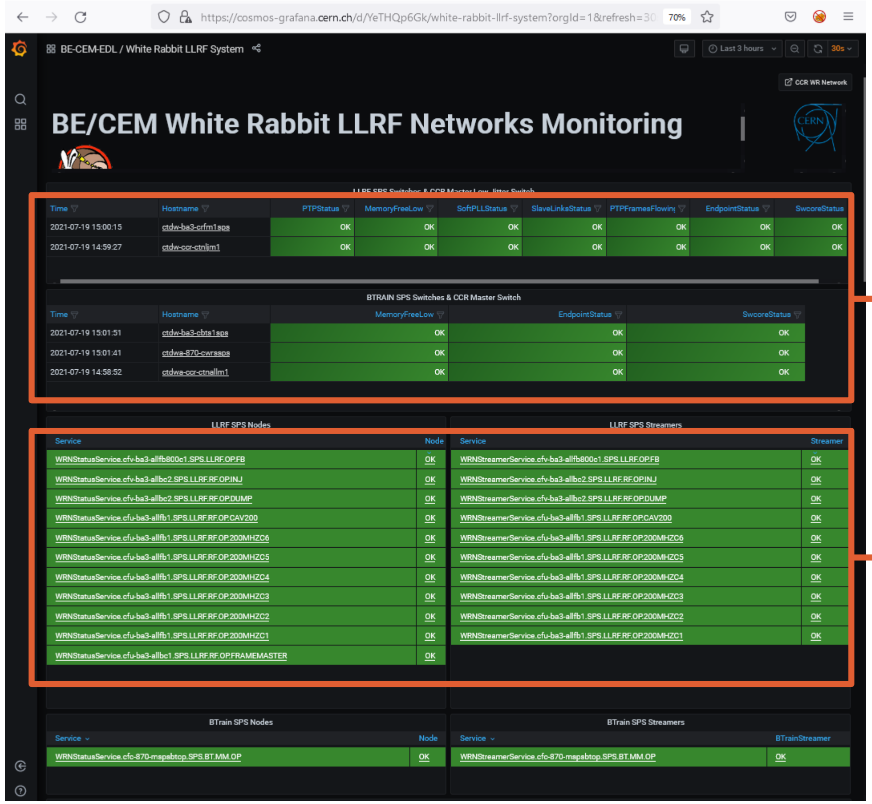
- **Example: PS – SPS synchro**



Revolution frequency distribution

- **Several equipment rely on the revolution frequency or a multiple of it**
  - RF Cavity controllers, kickers, injection/extraction synchro, beam instrumentation, …
- **It is distributed digitally over a white rabbit network (Frequency Tuning Word) and also in analog via copper/fibers**



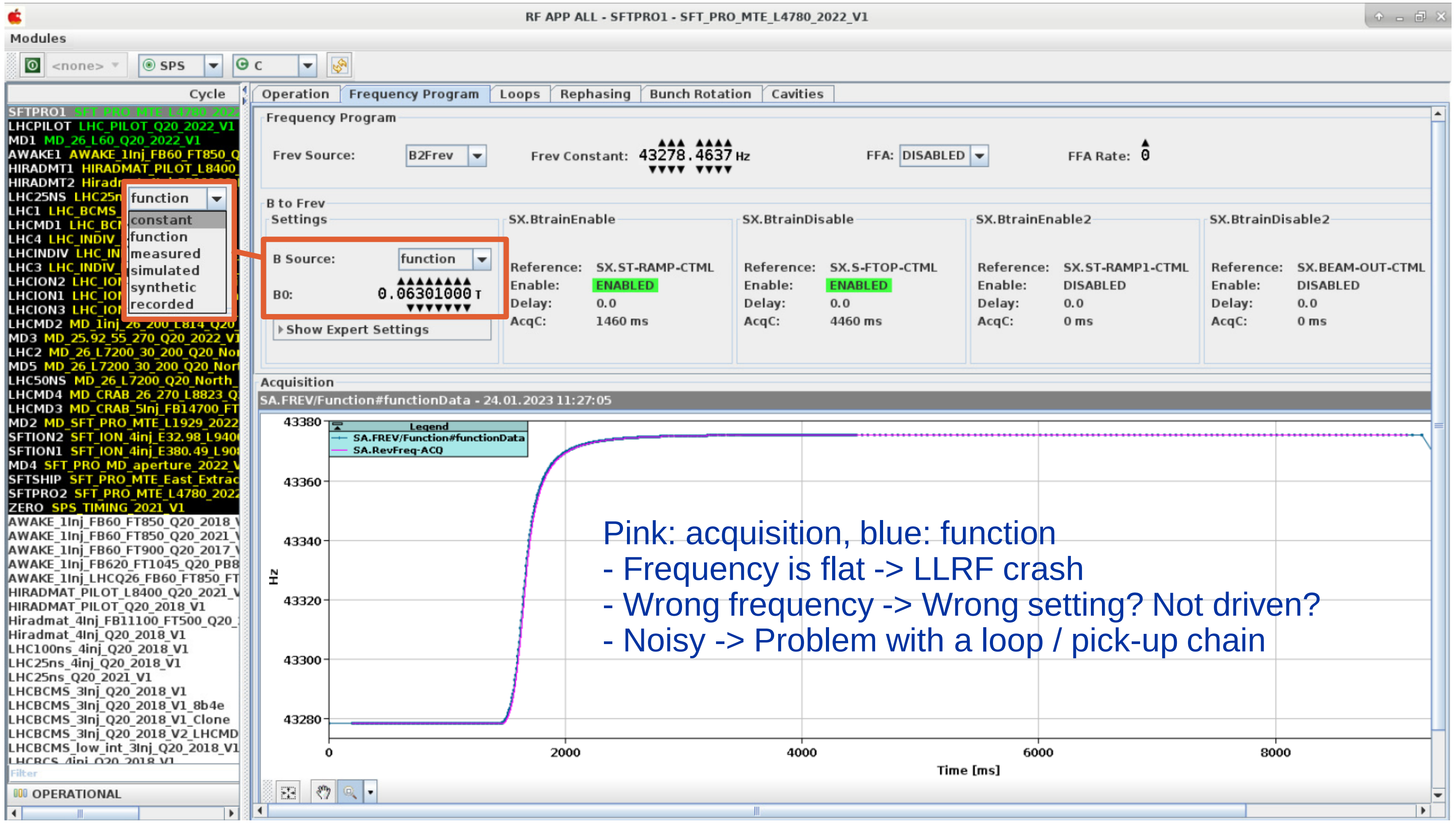


CCM -> White Rabbit LLRF Monitoring

Switches: WR support

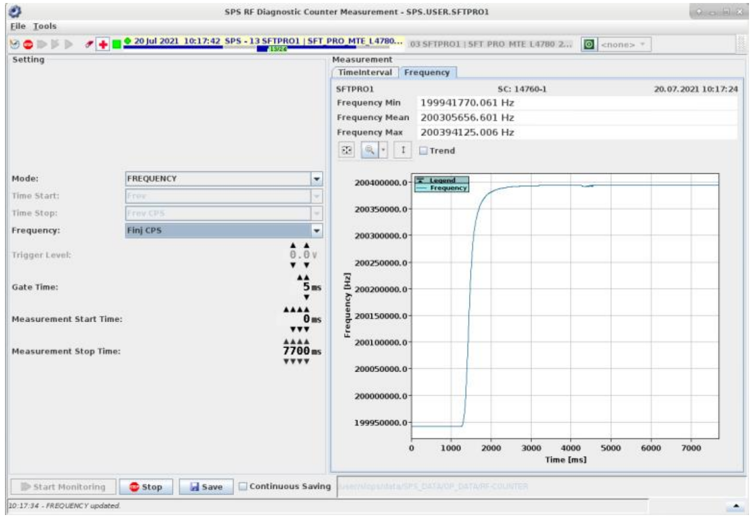
LLRF nodes: reboot/LLRF expert

Revolution frequency debugging

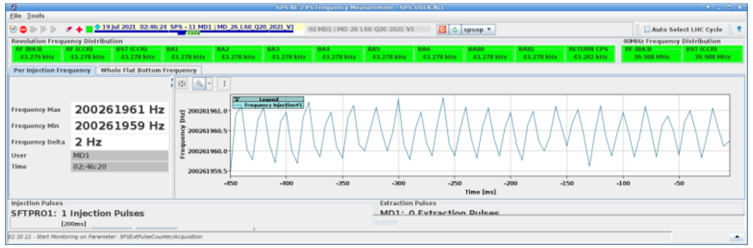


Pink: acquisition, blue: function
- Frequency is flat -> LLRF crash
- Wrong frequency -> Wrong setting? Not driven?
- Noisy -> Problem with a loop / pick-up chain

CCM -> RF APP ALL: Main entry point for LLRF Beam Control (F. Follin)



CCM -> Diagnostic counter:
Independent frequency measurement of
- Fc, Frf Ext (400 MHz rephasing)
- Frev to PS (Synchro)
- Frf to PS (Synchro)
- Frev Frf



CCM -> SPS 2 PS RF frequency Measurement:
Reports issue with the injection frequency

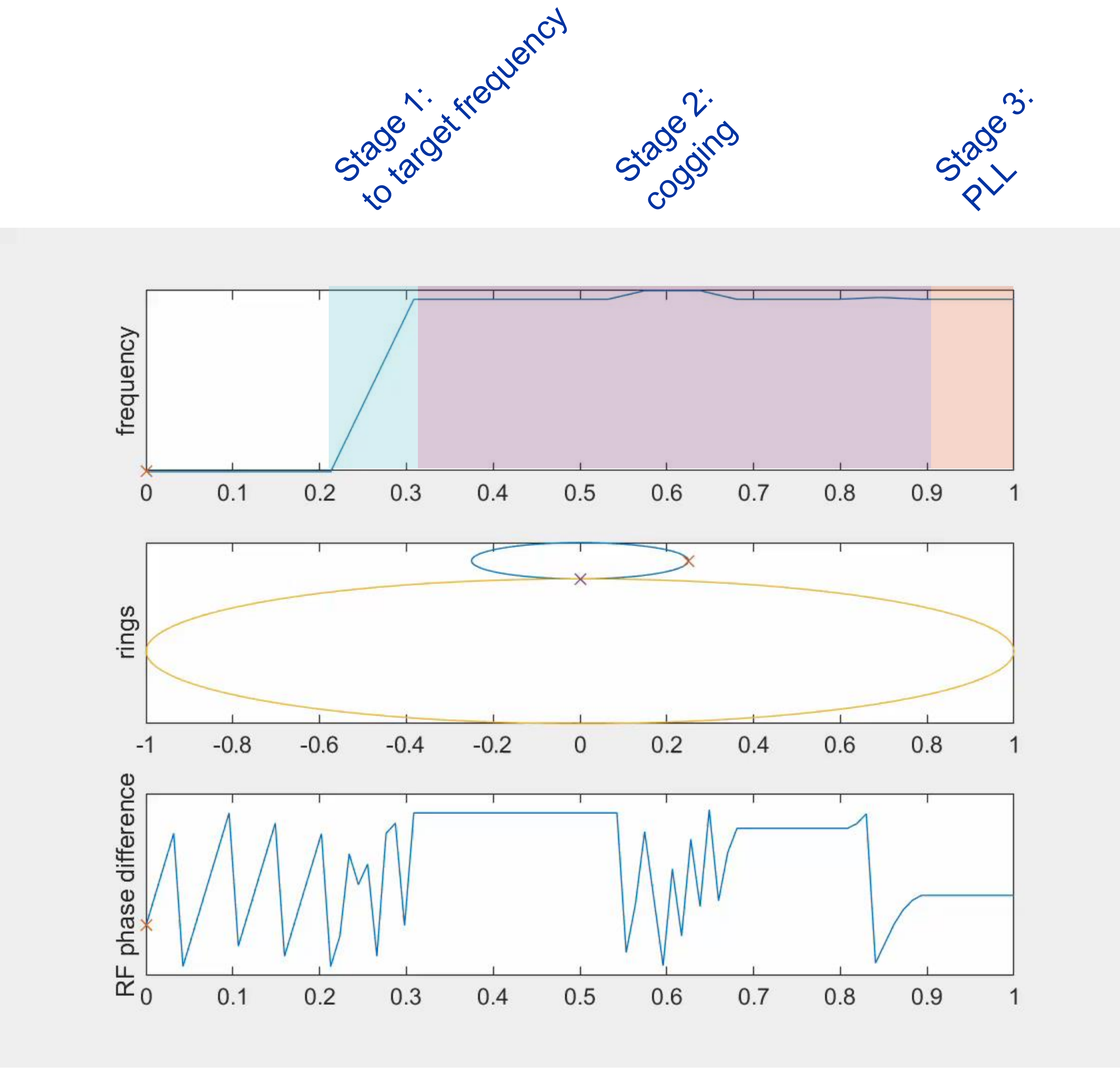
Rephasing

Rephasing principle

- **The rephasing is a synchronization sequence to rotate and extract the beam in the correct bucket (LHC, AWAKE)**

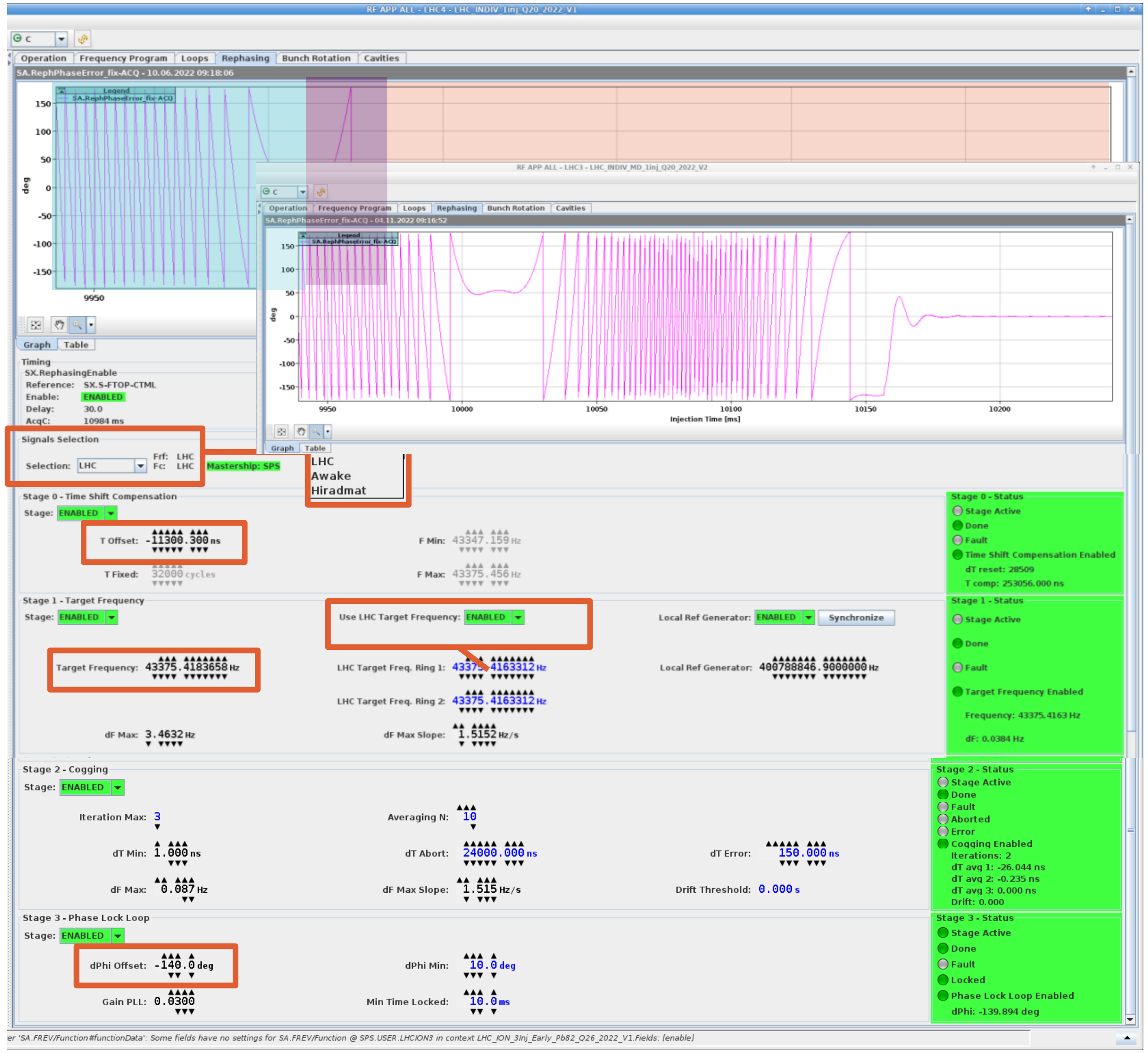
- **The reference is f_c , the common frequency between two machine (provided by downstream machine)**
  - For LHC, $f_c = frev_{LHC}/7 = frev_{SPS}/27 = \sim1.6$ kHz
  - For AWAKE, f_c is synchronized with the laser pulse

- **Once the first bunch is in the correct position the SPS RF is locked to the master machine RF with a feedback (PLL)**



Rephasing setting up

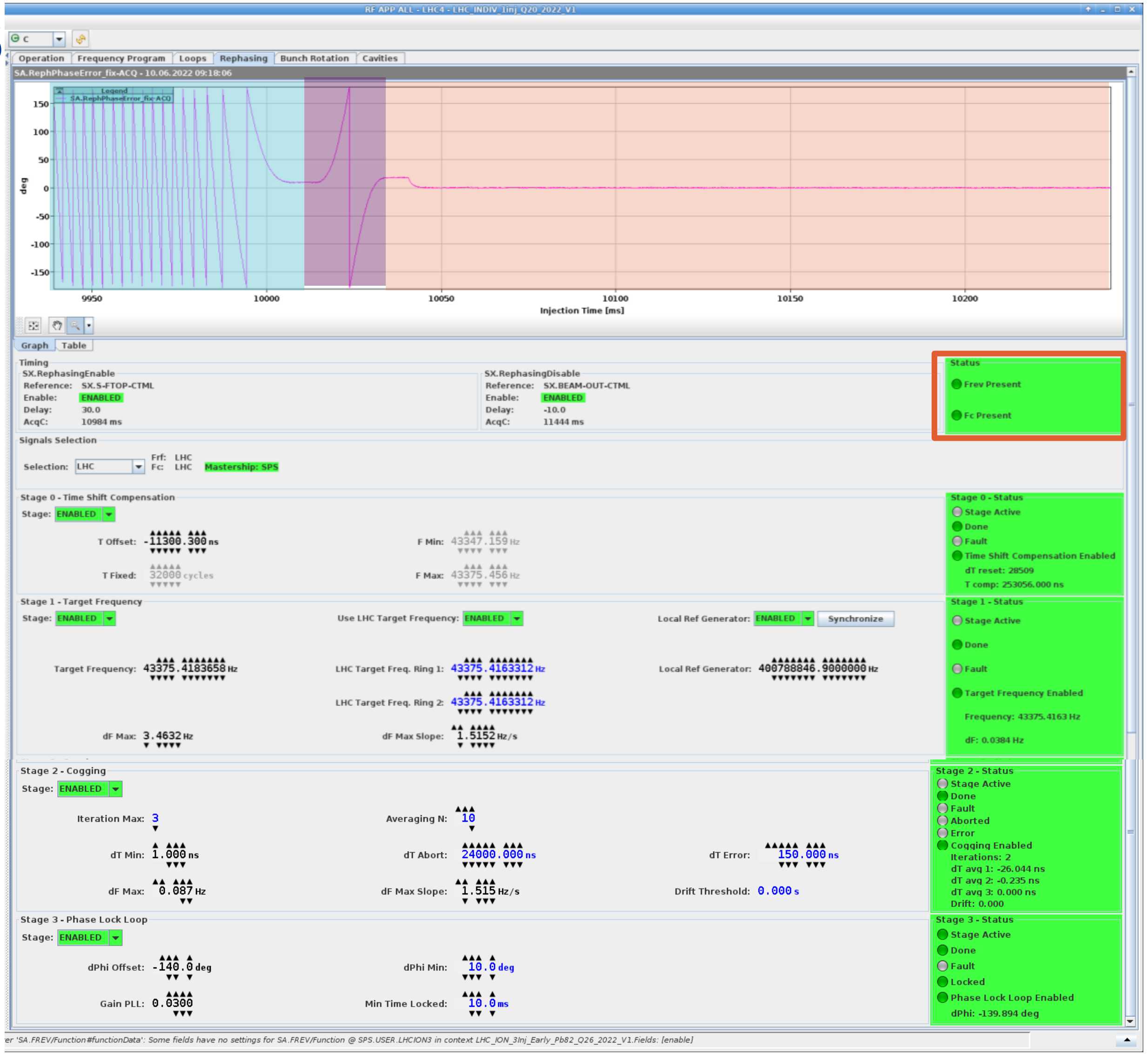
- **Setting the signal sources**
  - The rephasing can operate on LHC, AWAKE or a local RF/pulse generator during setting up
- **T offset (stage 0) is a delay used to adjust the position at extraction. It is adjusted with the BQM first bunch position as reference.**
  - With a fine adjustment, the phase when locking the PLL is brought towards zero to avoid a bucket jump.
- **The target frequency (stage 1) is set either to LHC (automatically tracking) or manually entered for other targets**
- **The cogging settings (stage 2) define the shape of the frequency bumps**
- **The phase lock loop (stage 3) dPhi offset is adjusted with the BQM fine bunch position as reference.**
- **If the sequence cannot execute withing the allocated time, stage 0 will be optimized at the next cycle**



CCM -> RF APP ALL: Rephasing tab

Rephasing common issues

- **Missing Fc**
  - Can be checked with diagnostic counter
  - If present and frequency correct -> LLRF issue
- **Wrong target frequency**
  - Phase is not flat during measurement
  - If PLL locks, can cause radial steering
- **Cogging fails**
  - Leave at least one training cycle
  - Check fc
- **A status will be added for frr.ext presence**



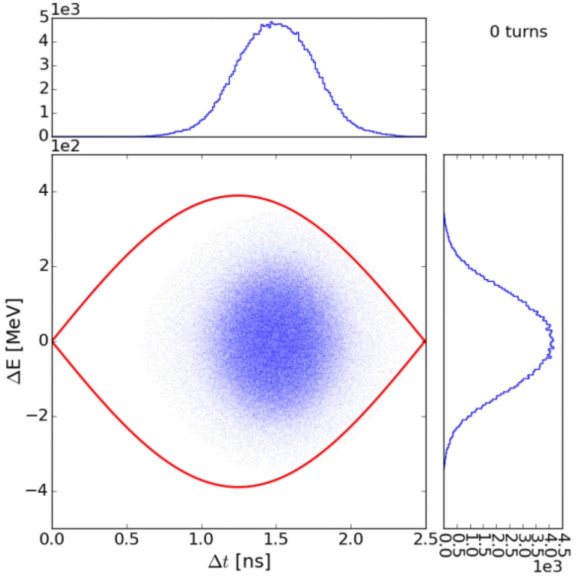
CCM -> RF APP ALL: Rephasing tab

Beam-based loops

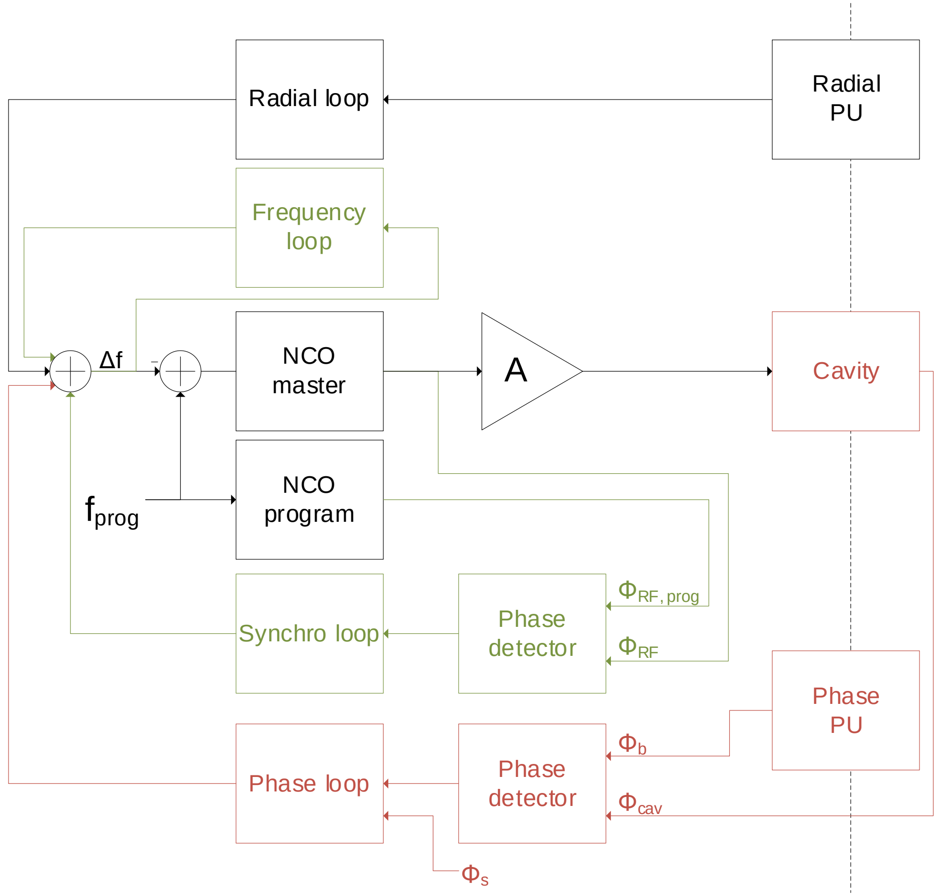
Two systems: phase + synchro loops or phase + radial loops

Phase and synchro loop, what for

- **The Phase loop reduces the jitter between beam phase and Cavity Field**
  - It damps the injection phase/energy errors
  - It reduces the effect of RF noise (longitudinal emittance growth)
- **The Synchro loop locks the RF onto an phase reference (program)**
  - It is essential at injection (bunch into bucket transfer)
  - It can be used in the ramp if the reference RF follows the momentum ramp
  - It is essential at extraction (locks the SPS buckets onto the LHC or AWAKE reference)



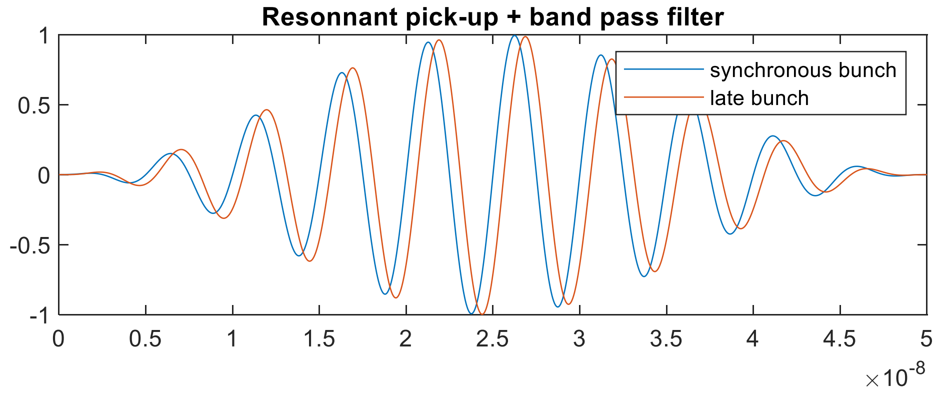
Phase space evolution:
phase error at injection
Analysis by H. Timko

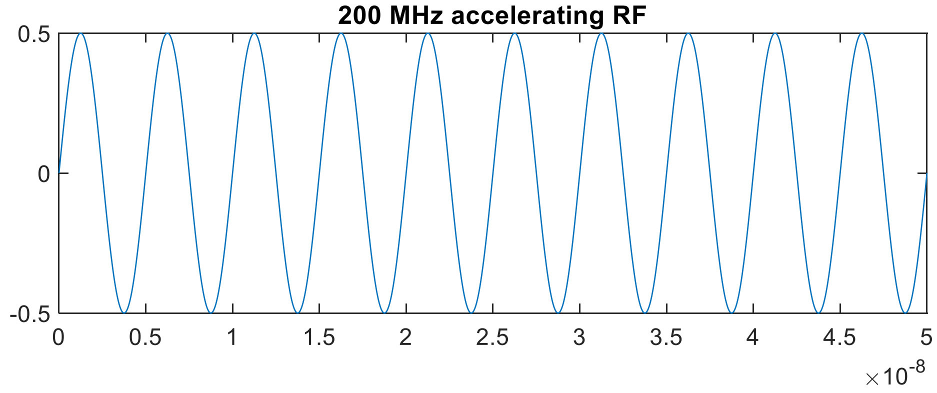


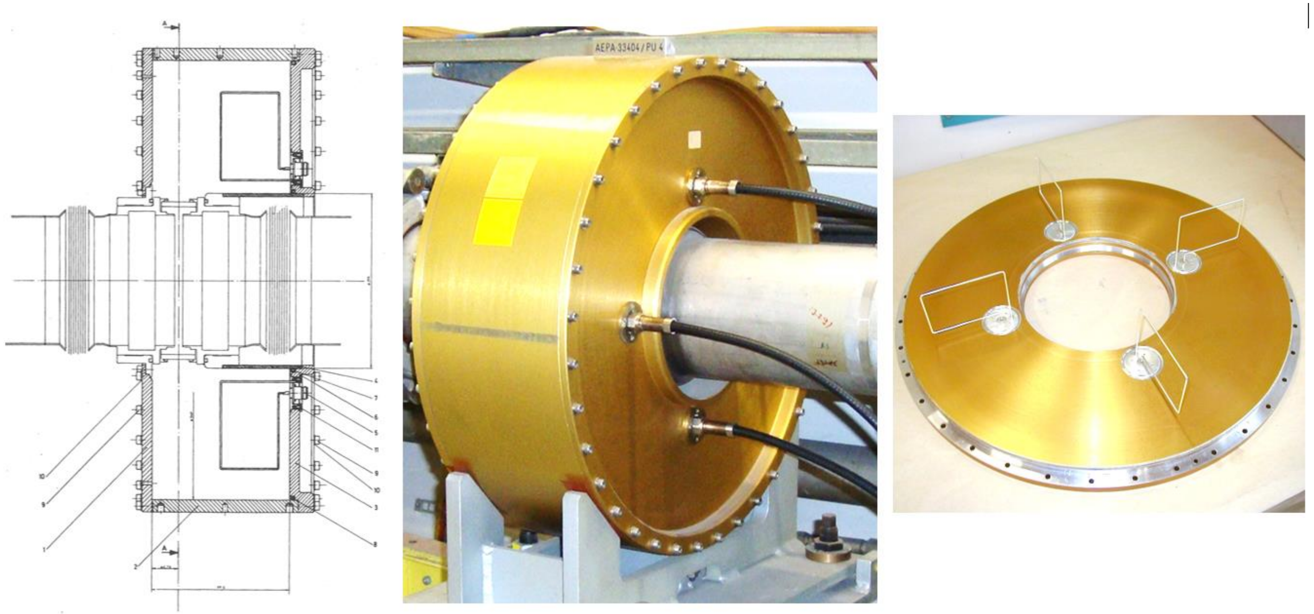
Narrow band Phase pickup principle

- The passing bunch induces a resonance in the cavity, its centre frequency is 200 MHz.

- After filtering (band pass) and gain adjustment, we can compare it to the accelerating RF and extract the phase





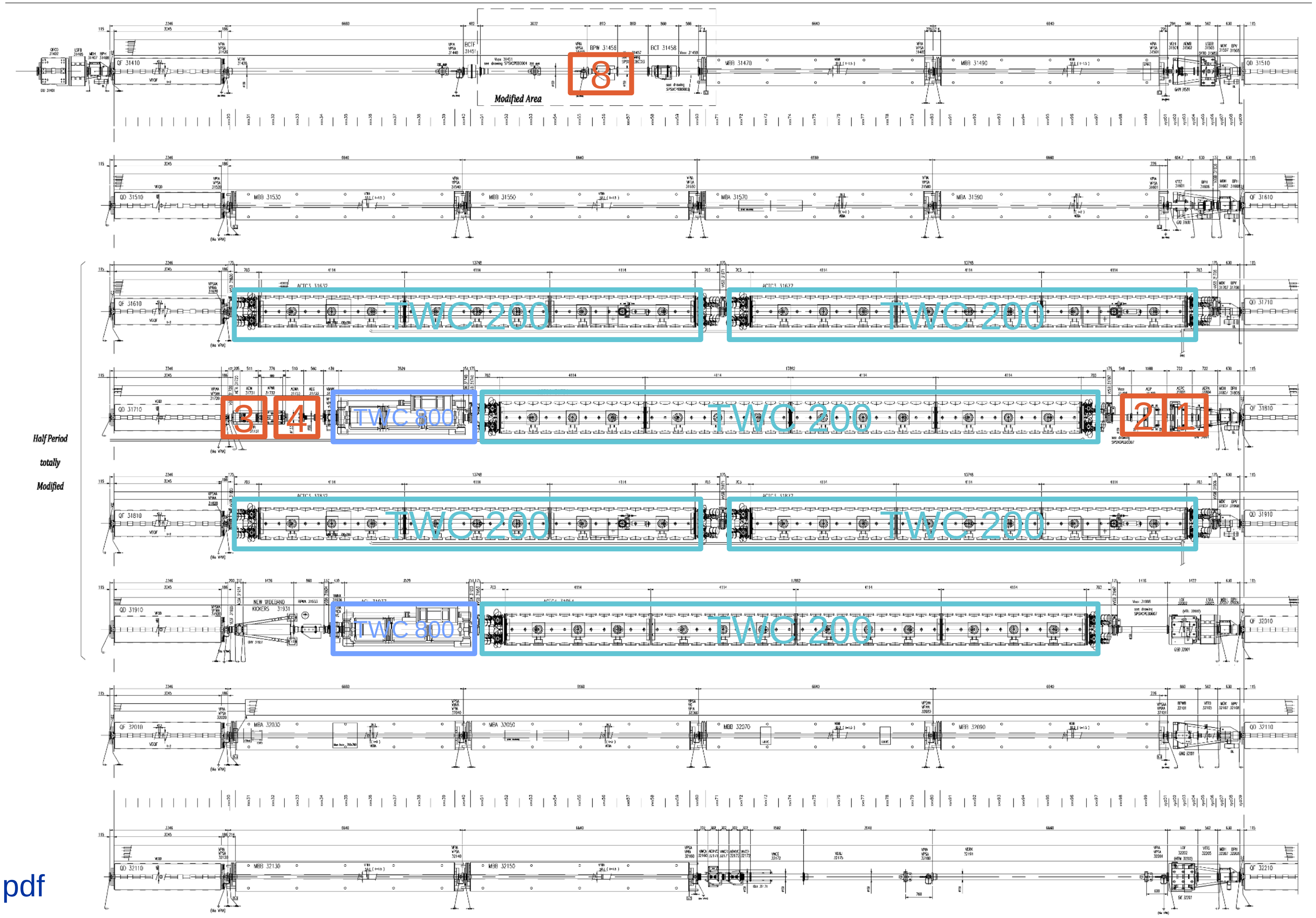


a) Schematic design b) Assembly in the tunnel c) Coupling loops

Resonant phase pickup
Picture courtesy of xxx

Where are the pick-ups

- **Longitudinal Narrow band (resonant)**
  - AEPC.31801 (1)
  - AEP.31799 (2)
- **Longitudinal Wide band (wall current)**
  - AEW.31731 (3)
  - AEWA.31733 (4)
- **Transverse Narrow band**
  - BPCR.31202 (5)
  - AERB.31302 (6)
- **Transvers Wide band**
  - BPWA.31101 (7)
  - BPW.31456 (8)

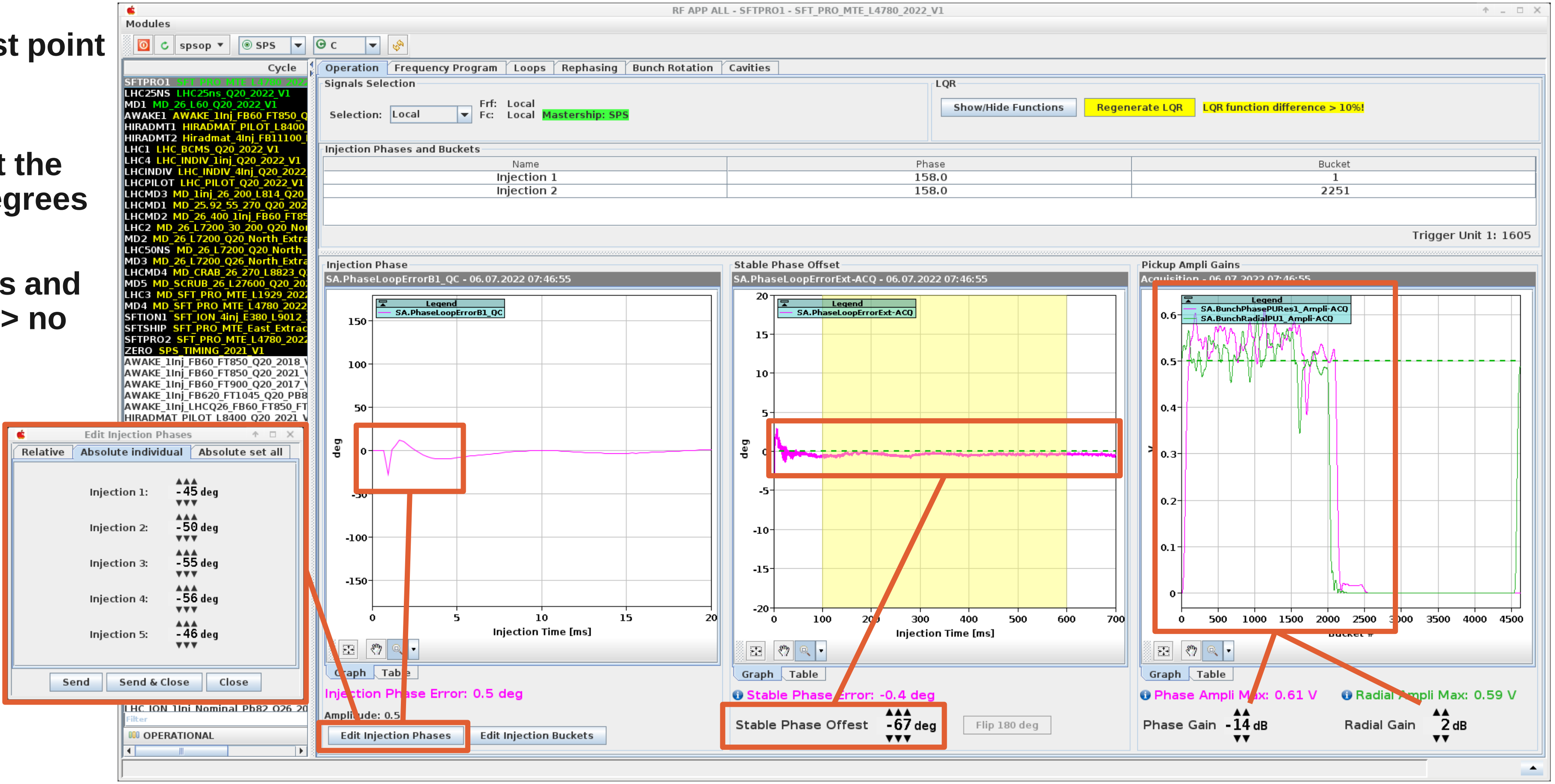


LSS3 arrangement
edms.cern.ch/ui/file/1155058/AB/spslnins0106-vAB.pdf

Injection phase

- Adjust pick-up gain to have ~0.5 peak amplitude (0.25 for slip-stacking)
- Adjust injection phase to get first point around zero degrees
- Adjust stable phase offset to get the injection plateau around zero degrees
- Warning: if the beam de-bunches and intensity goes below threshold -> no more acquisition
  - Maybe flip 180 degrees
- For high intensity beams, the feed forward will prevent over-compensation

CCM -> RF APP ALL: Operation tab

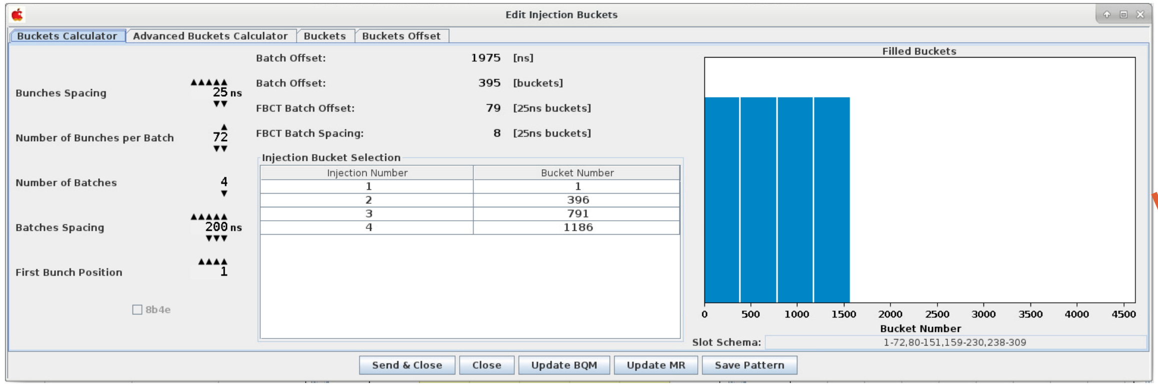


Masks

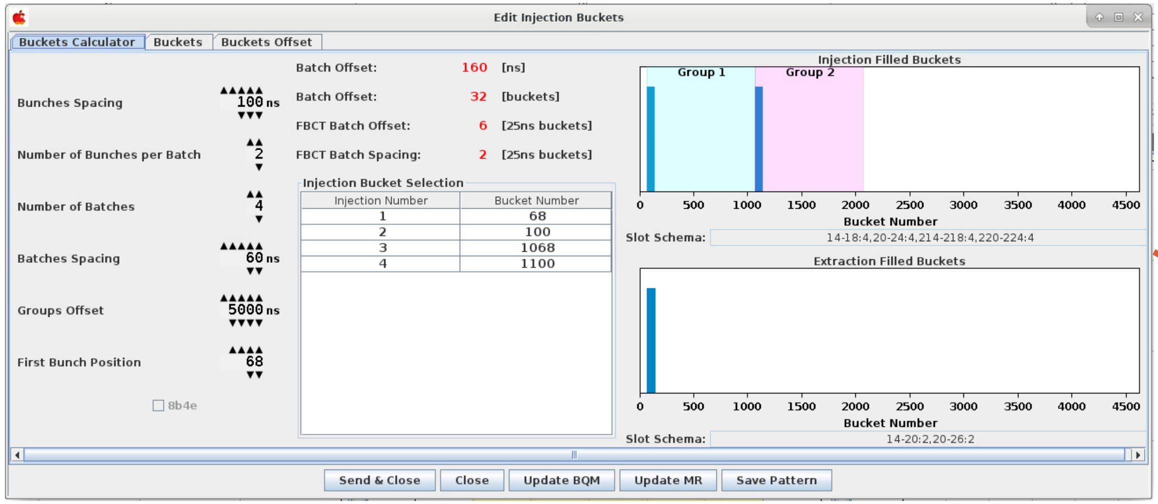
- **Injection buckets setup**
  - Used by loops masks
  - BQM
  - Mountain range

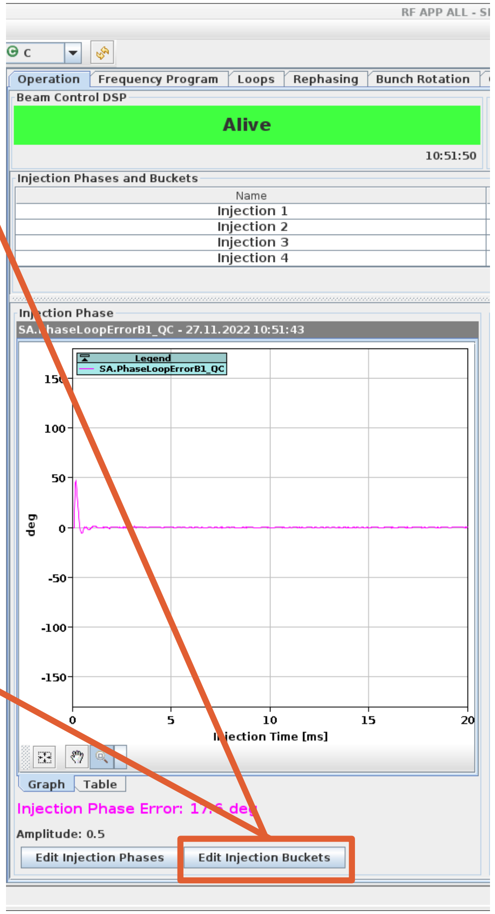
- **For ions slip stacking, the additional group offset attributes bunches to the two phase loop**

LHC25ns



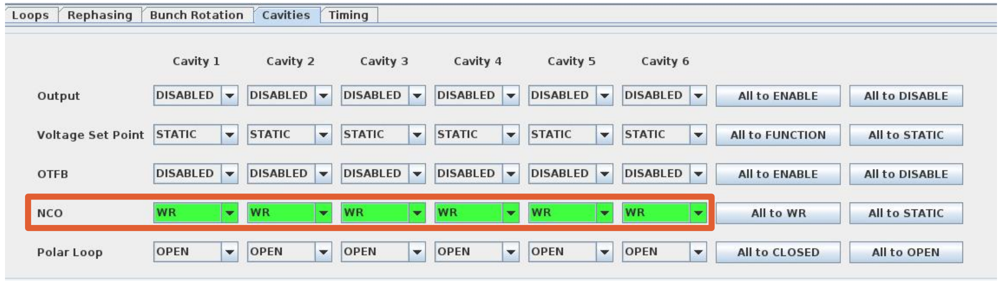
LHCION1



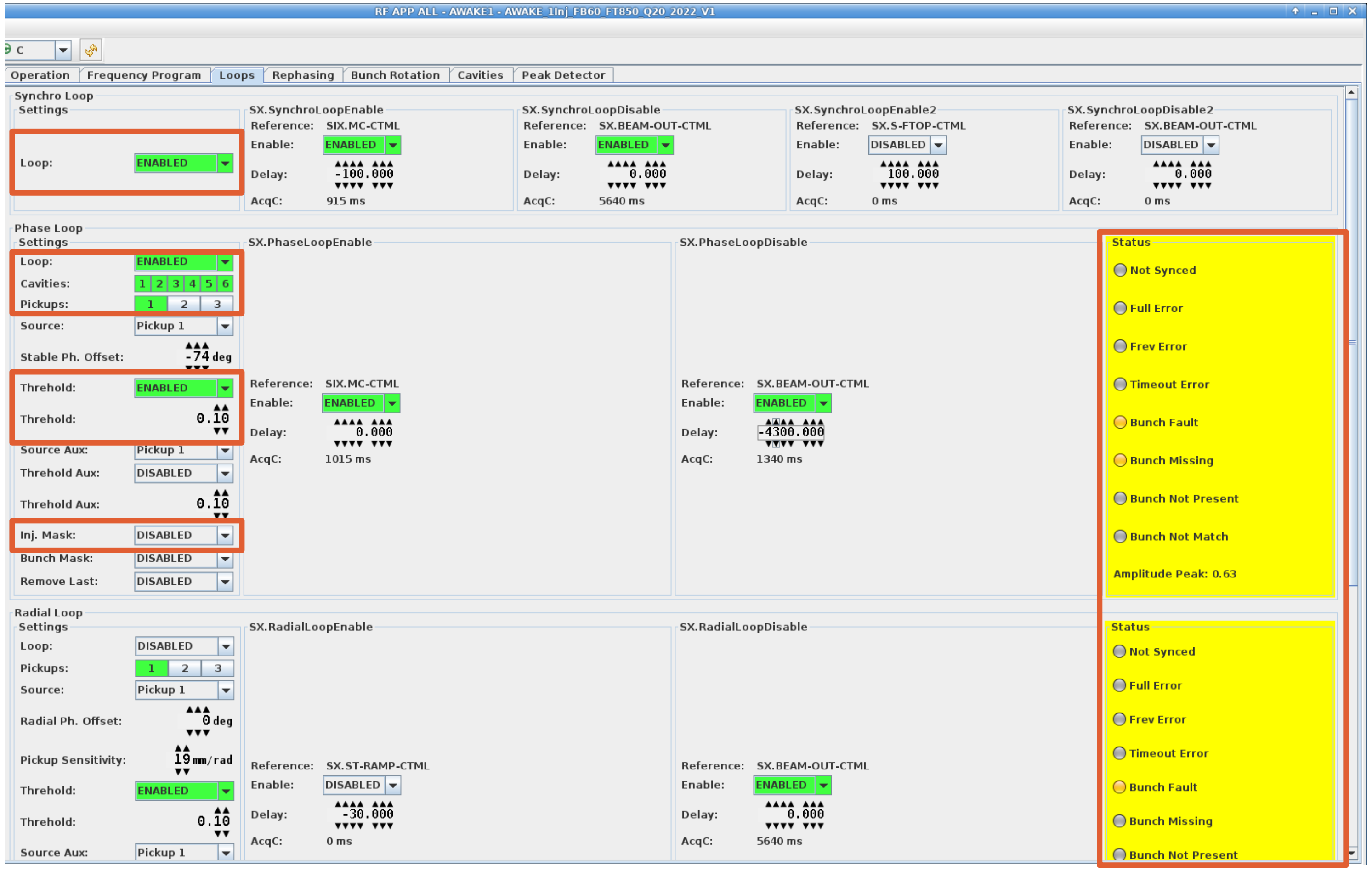


Status and controls

- Enable/disable loops
- Setup cavities/pickup sources
- Setup masking and bunch detection threshold
- To monitor the status : If boxes go red, it is likely a synchronization issue with a cavity controller (It might be set on static frequency or crashed)



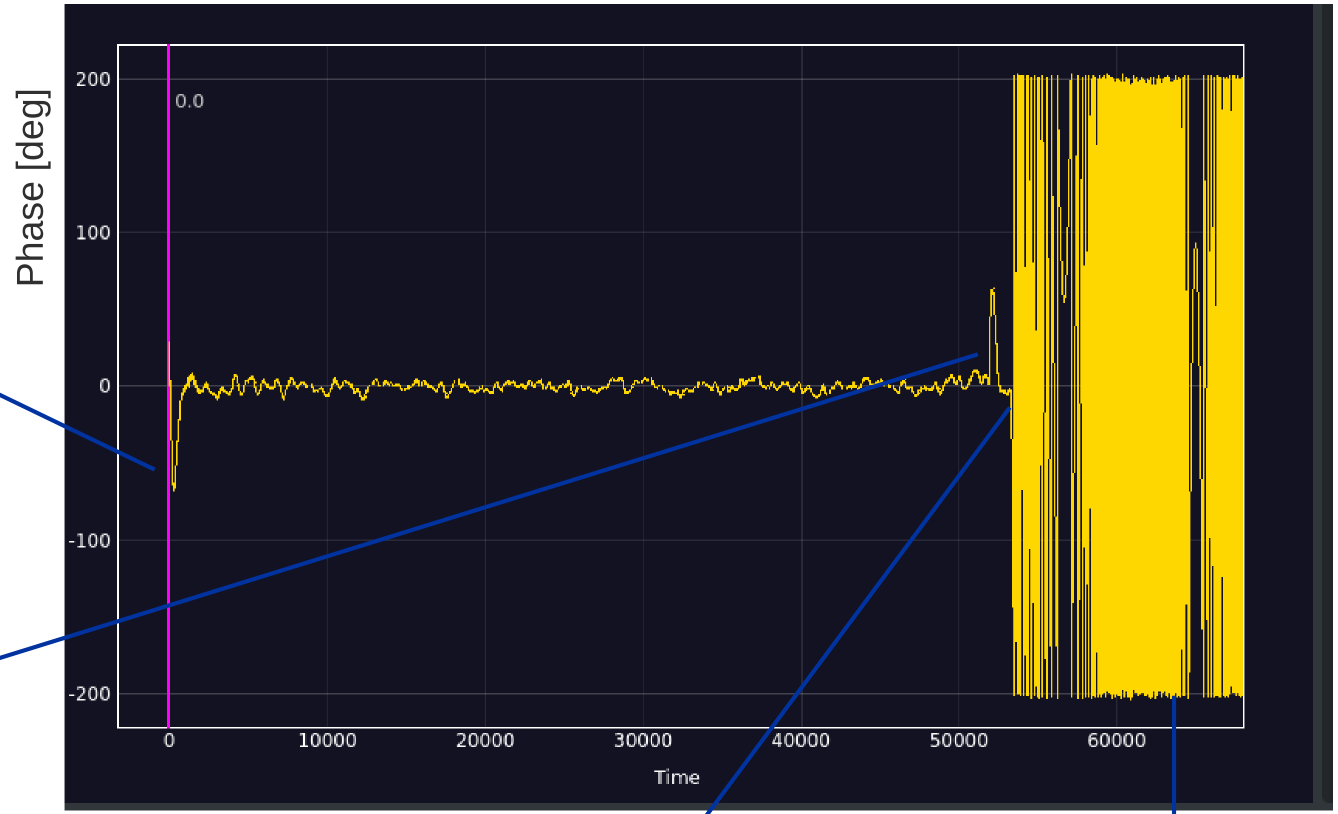
CCM -> RF APP ALL: Cavities tab



CCM -> RF APP ALL: Loops tab

Beam loops: Synchro loop error (SFTPRO)

CCM -> Megadrive: Synchro loop error



Injection batch 1
Energy mismatch

Injection batch 2
Some difference in energy

Switch Synchro-Radial loops

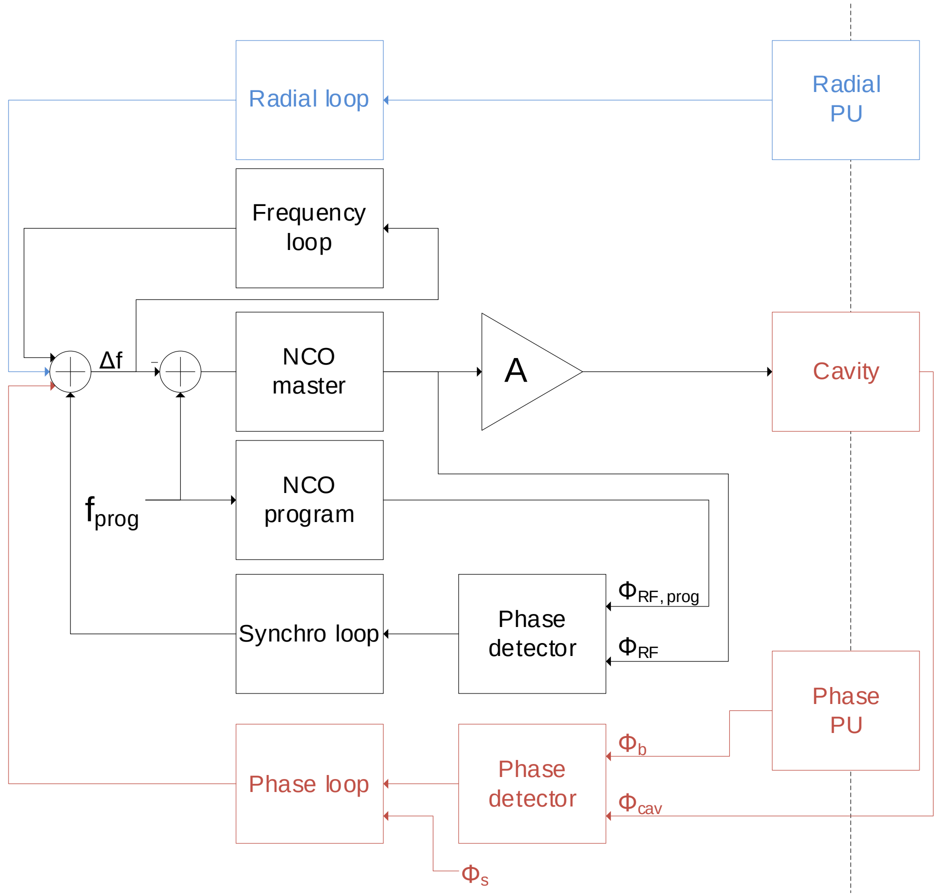
Ramp with radial loop.
Synchro loop unlocks

phase and radial loop, what for

- **The Phase loop reduces the jitter between beam phase and Cavity Field**

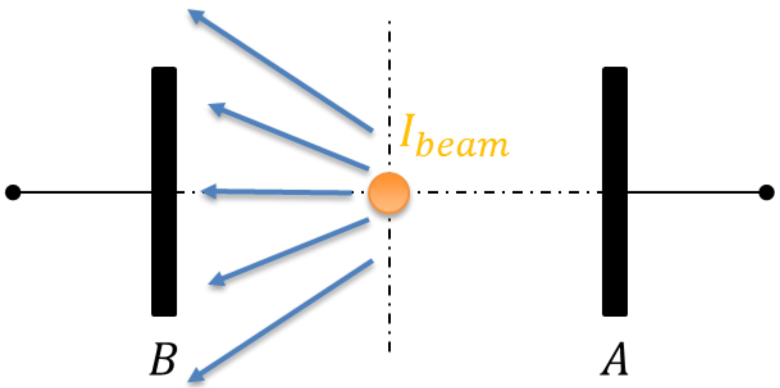
- **The Radial loop keeps the mean orbit centred**
  - It is essential to cross transition (where a small RF frequency error results in very large orbit displacement) -> control of frequency is not sufficient
  - It cannot be used at injection as it does not guarantee constant RF frequency (influence of the errors in radial measurements)

- **Cycles**
  - SFTPRO
  - SFTION, LHCION

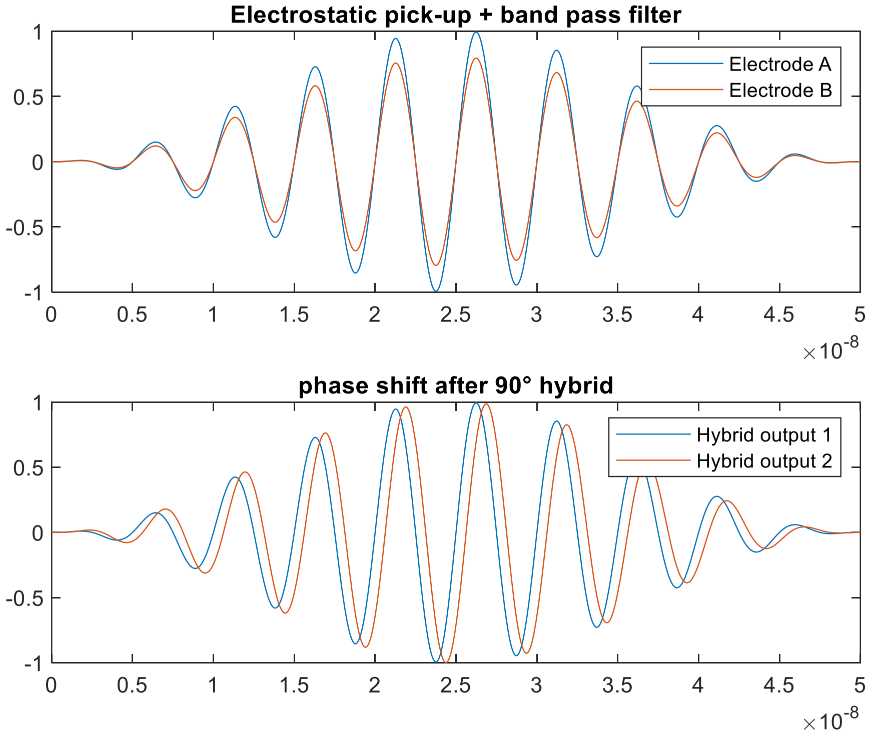


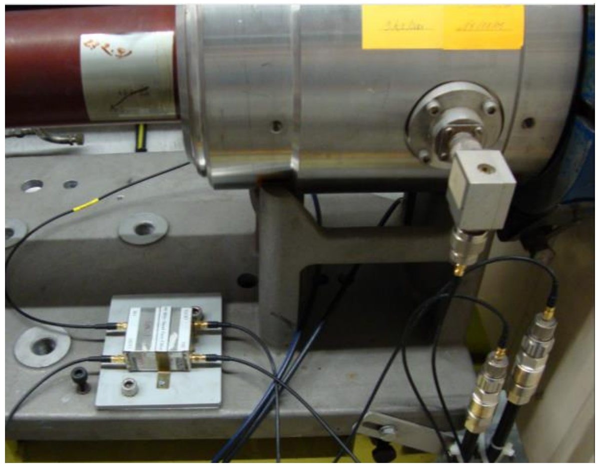
Radial pick-up principle

- Two opposing electrodes, A and B are sensing the beam induced electro-magnetic field
- The electrode voltage is fed in a 90° hybrid, where the amplitude difference due to beam position is translated into a phase difference.
- The radial position is ~proportional to the extracted phase



Radial pick-up electrodes A and B
Drawing courtesy of G. Kotzian

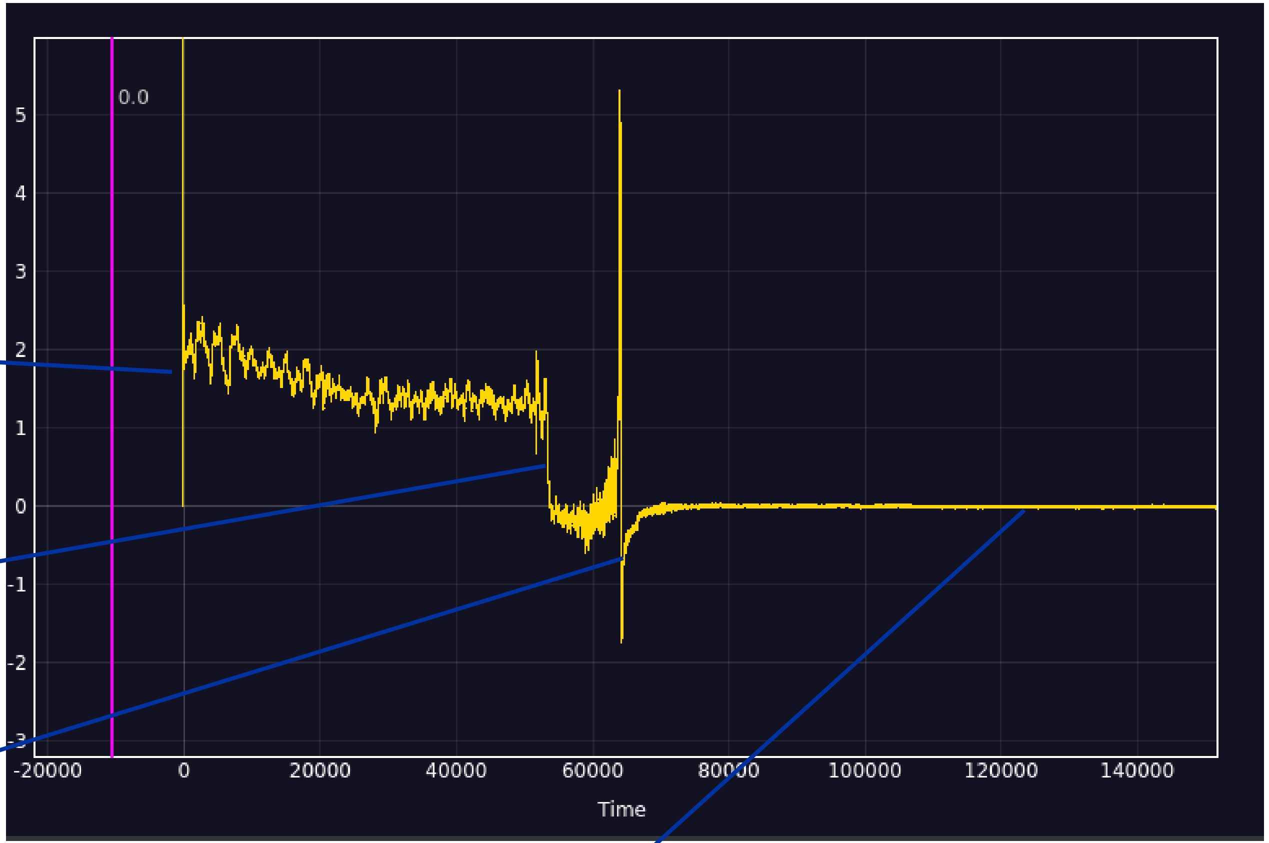




AERB.31302 electrostatic pick-up
Picture courtesy of T. Bohl

Radial loop error signal

CCM -> Mega Drive: radial loop error



Filling with Synchro loop: Radial position is not regulated well (B field decay)

Switch from Synchro to Radial loop: Radial position regulated to zero

Transition crossing

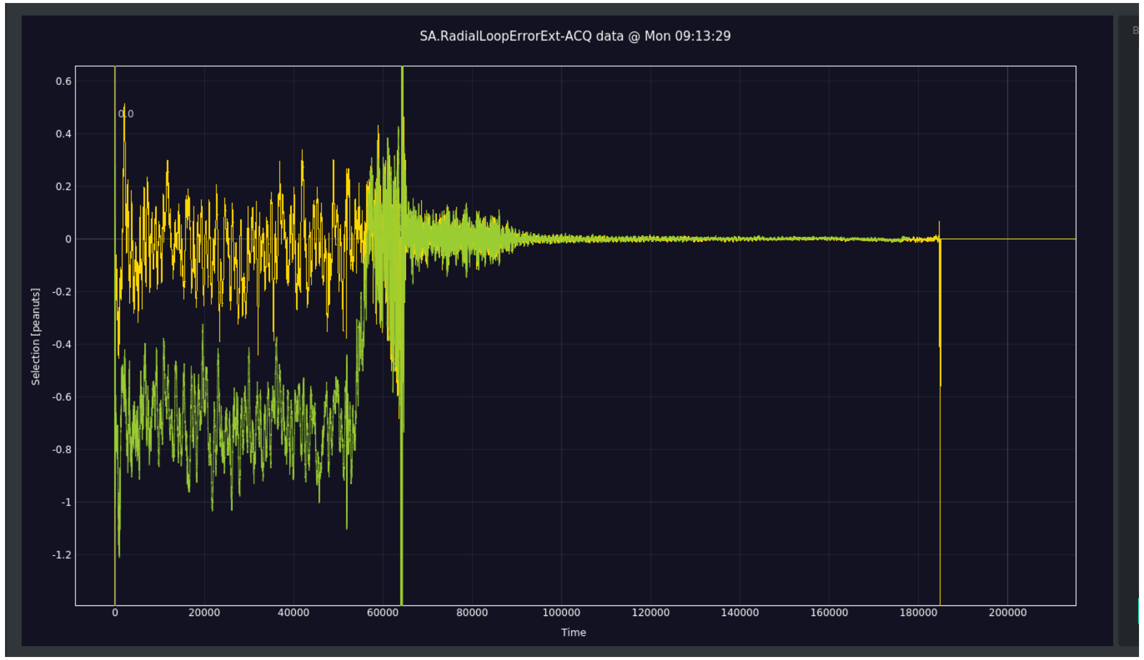
Radial position regulated at zero during ramp and flat top. Important for slow extraction

Radial loop calibration

- **Radial calibration against mean orbit at flat bottom (YASP)**
- **Done only a beam commissioning, unless it drifts…**

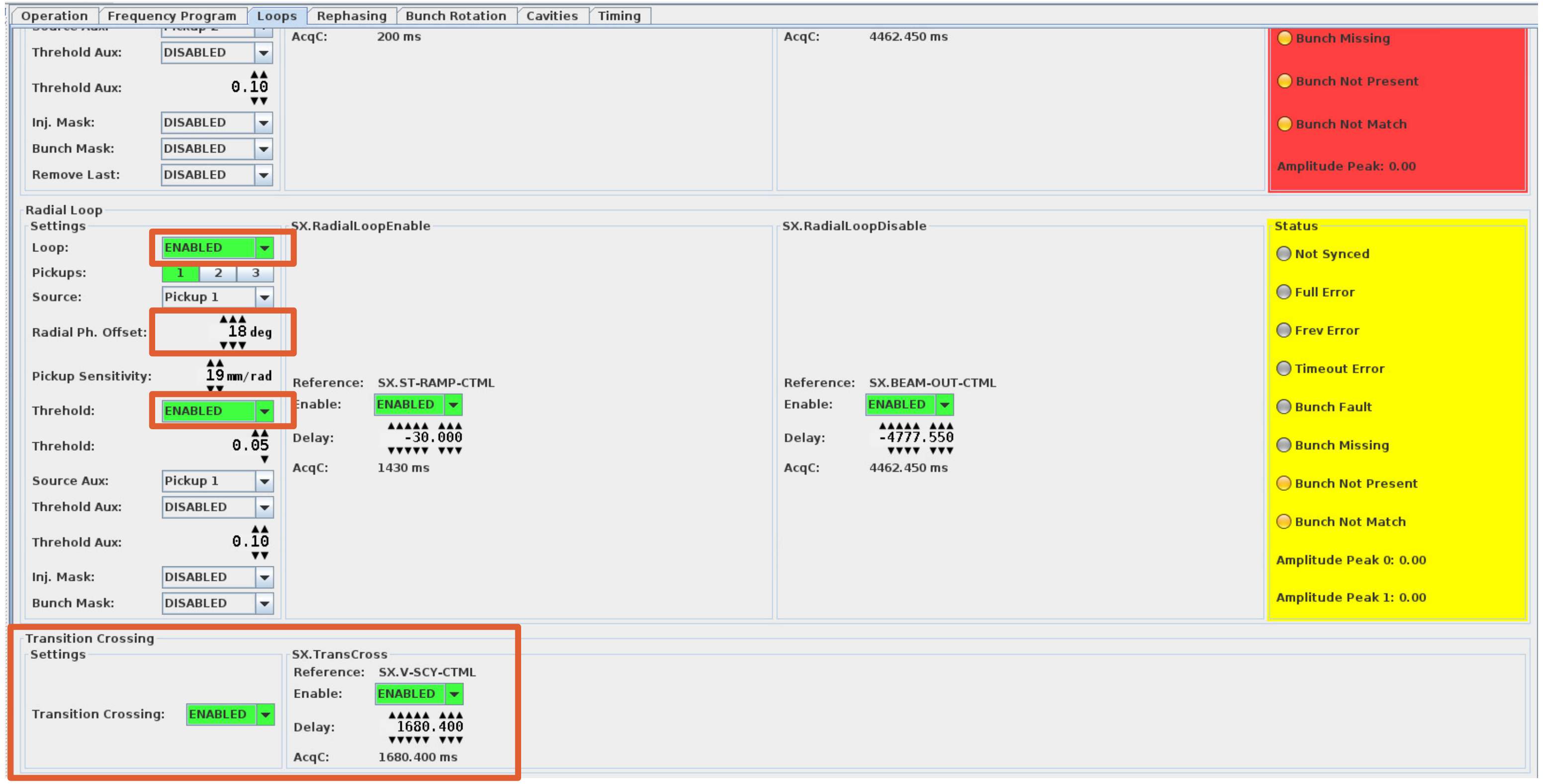
Calibrated (yellow)

Un-calibrated (green)



Radial loop controls

- **Enable/disable**
- **Calibration offset**
  - ~0.16 mm/deg
- **Threshold**
- **Transition crossing**
  - Flip the phase of 180°
  - Change sign of radial loop



CCM -> RF APP ALL: Loops tab

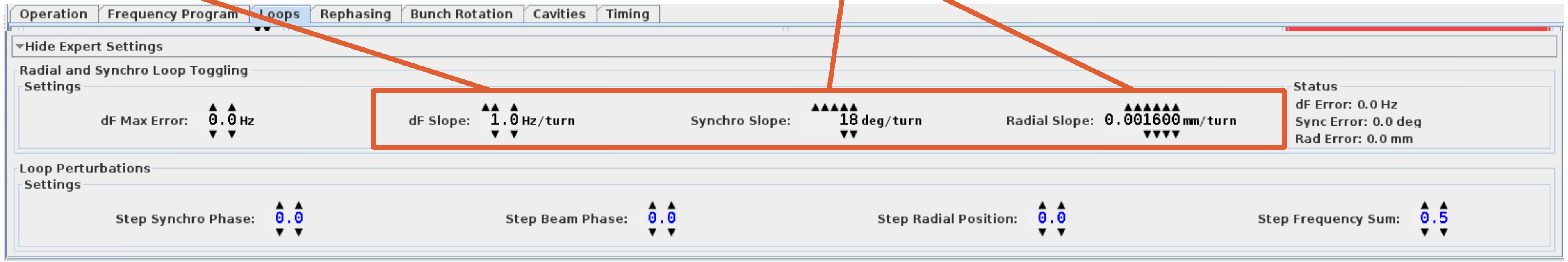
Loops switching: Synchro to radial loop

- **FT protons, LHC and FT Ions**
  - Automatically when radial loop enable timing occurs
  - Switches immediately without transient by “hiding” the potential radial error with an offset
  - Slowly ramp down the offset

- Generated settings suit all cases
  - If zero: an offset is kept during all the cycle
  - If too big: Causes RF phase transients

Decay speed of the frequency contribution of the loop that is switched off

Decay speed of the loop error offset at the moment of switching



CCM -> RF APP ALL: Loops tab bottom hidden expert setting !

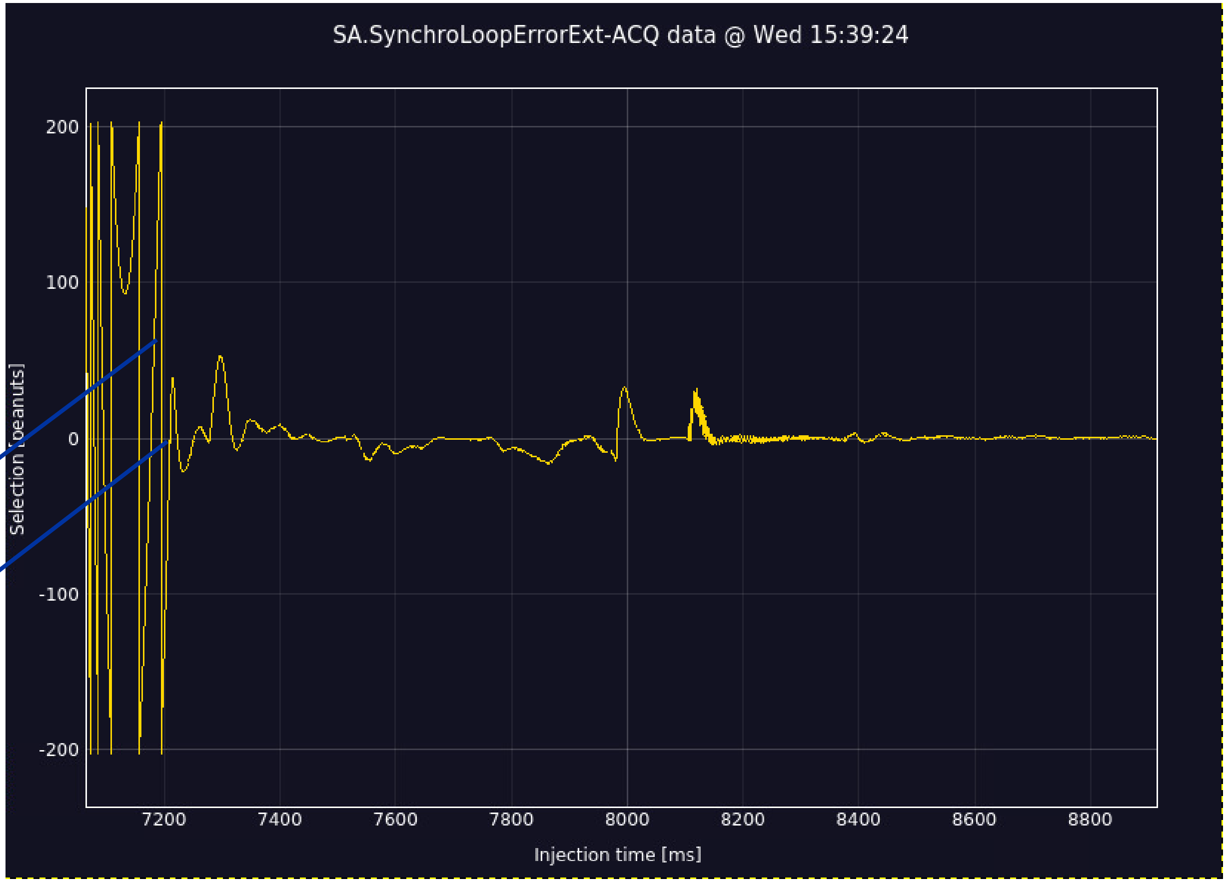
Loops switching: Radial to synchro loop

- **From radial to synchro loop (LHC Ions before rephasing)**
  - Automatically when synchro loop enable timing occurs
  - Turns radial loop off and ramp to program frequency + “beating offset”
  - Wait for zero crossing of the synchro loop error
  - Switch and ramp down beating offset

Replaces the need for a radial bump and several timings

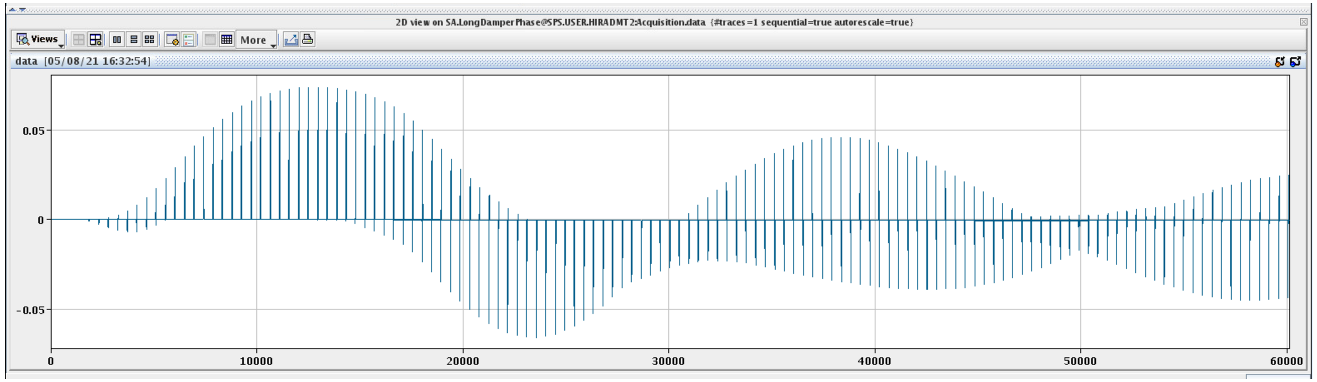
Beating frequency (program vs. current phase)

Zero-crossing: synchro loop closes

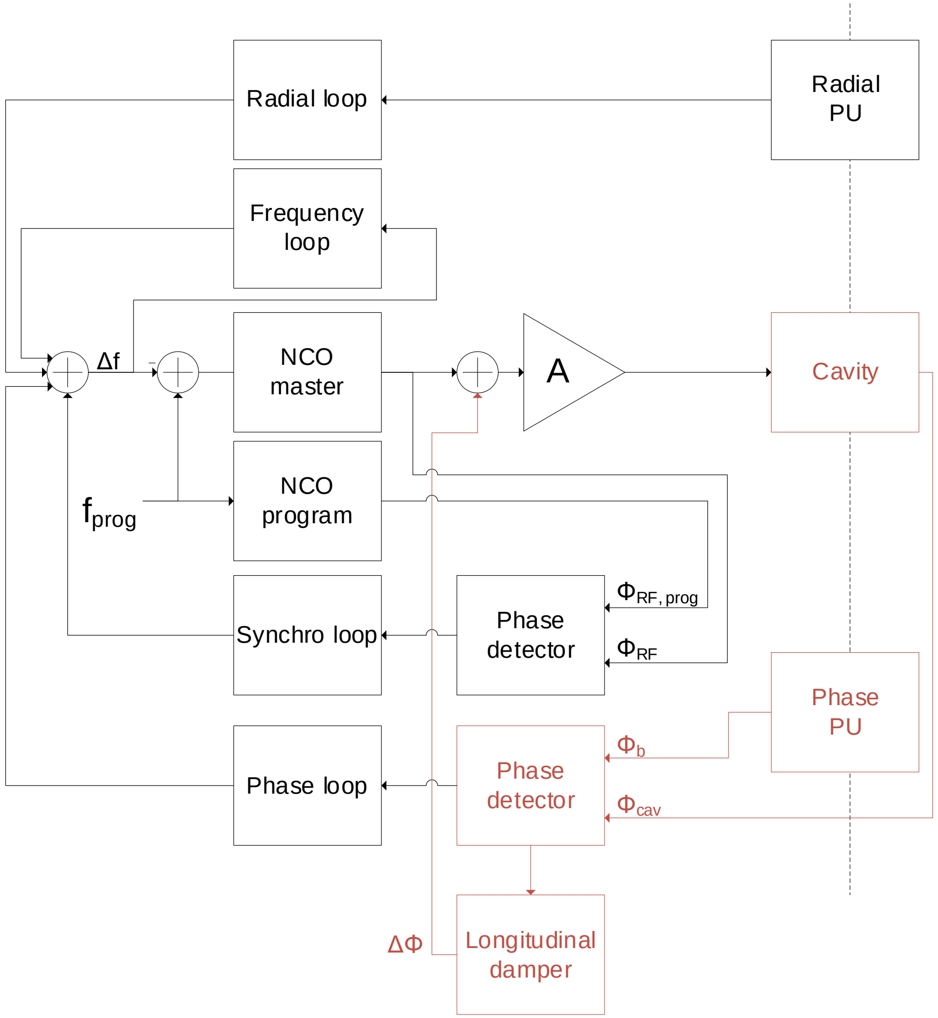


Longitudinal Damper principle

- **Bunch by bunch feedback to prevent longitudinal Coupled-Bunch Instabilities**
- **Modulates the phase of the 200 MHz RF cavities**
- **Damps last injection oscillations without “shaking” the circulating beam using masks for the phase loop**
- **Cycles**
  - High intensity LHC
  - AWAKE 3e11 ppb

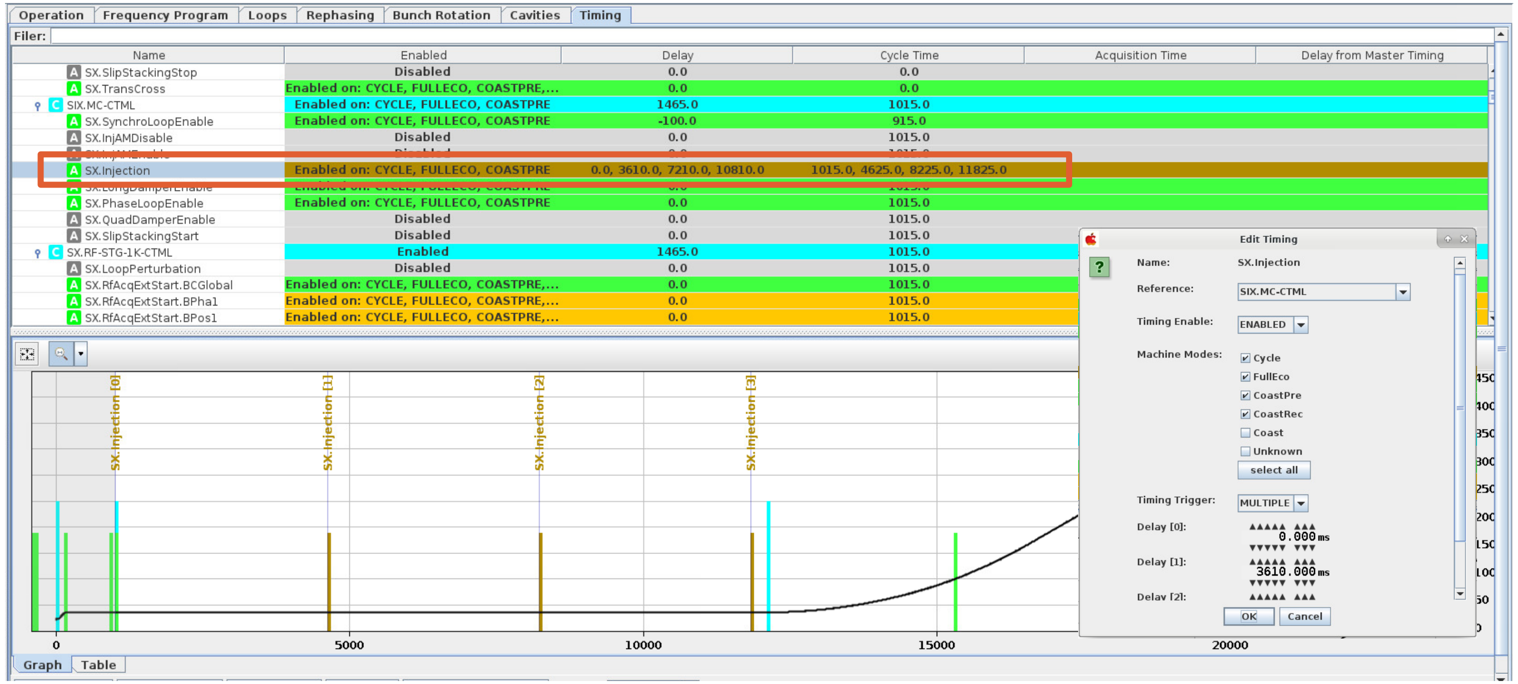


Phase kicks from the longitudinal damper to a single bunch

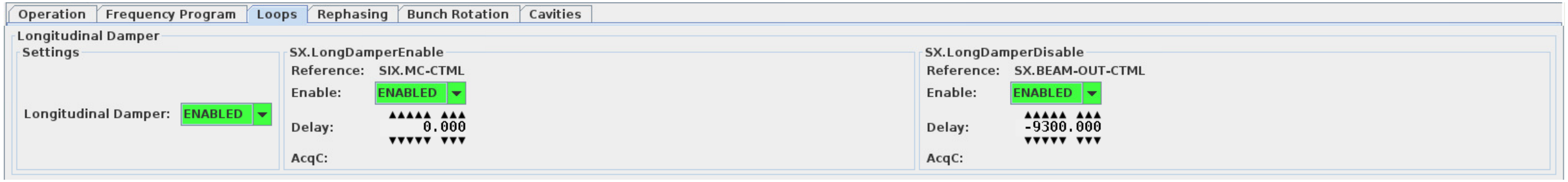


Longitudinal Damper setting up

- **The timings must be setup**
  - Start and stop (usually stopped beginning of the ramp when controlled blow-up starts)
  - Injections
- **The masks (filling pattern) should correspond to the beam**
- **It must be enabled in the cavity controllers**



CCM -> RF APP ALL: timing tab

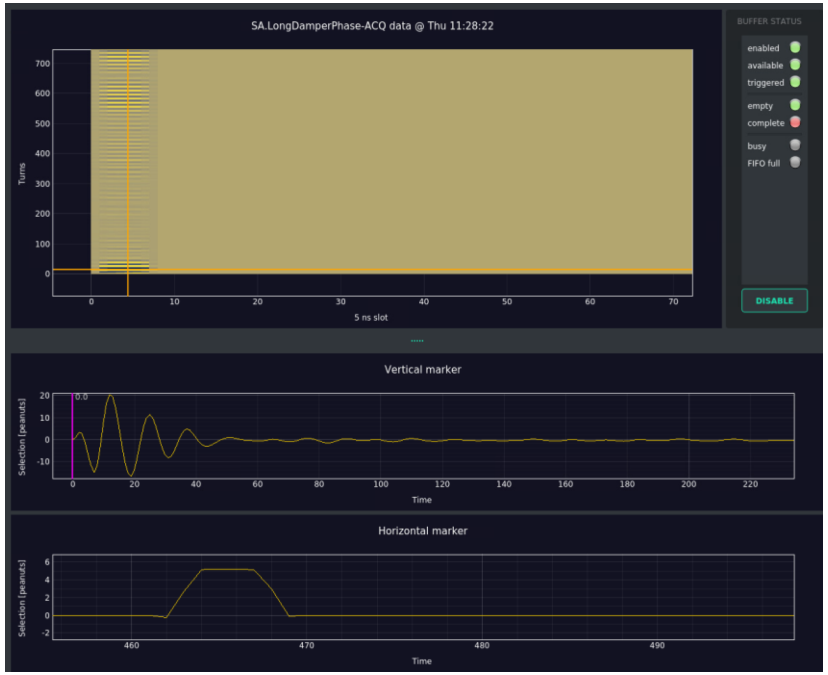


CCM -> RF APP ALL: Bottom of loops tab

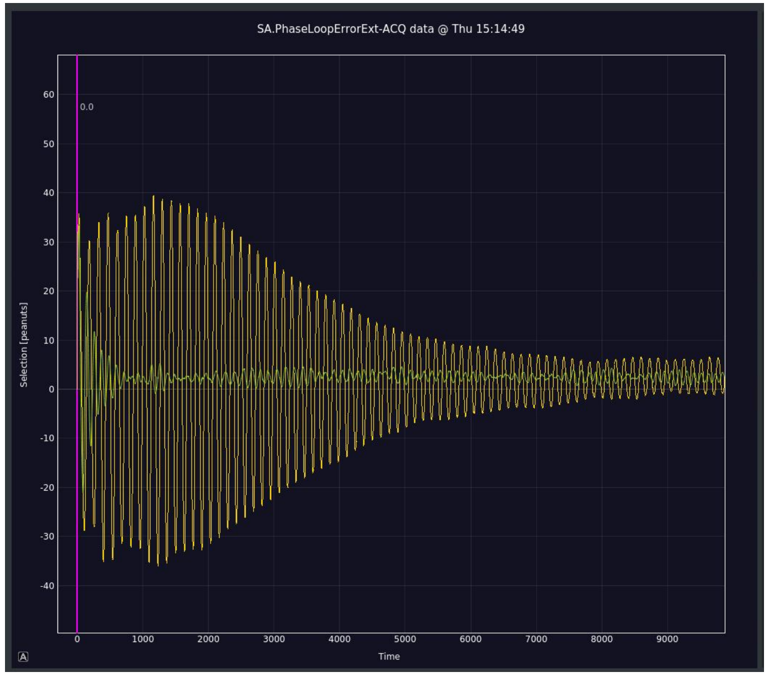
Longitudinal Damper setting up

- **Observation**

- **If unstable: The LD tracks the synchrotron tune, the calculated one (LSA function) should be within 10% of the real one.**



CCM -> Mega Drive: longitudinal damper phase



CCM -> Mega Drive: Damps injection oscillations but slower than phase loop (Green trace LD active)

Beam-based loops summary

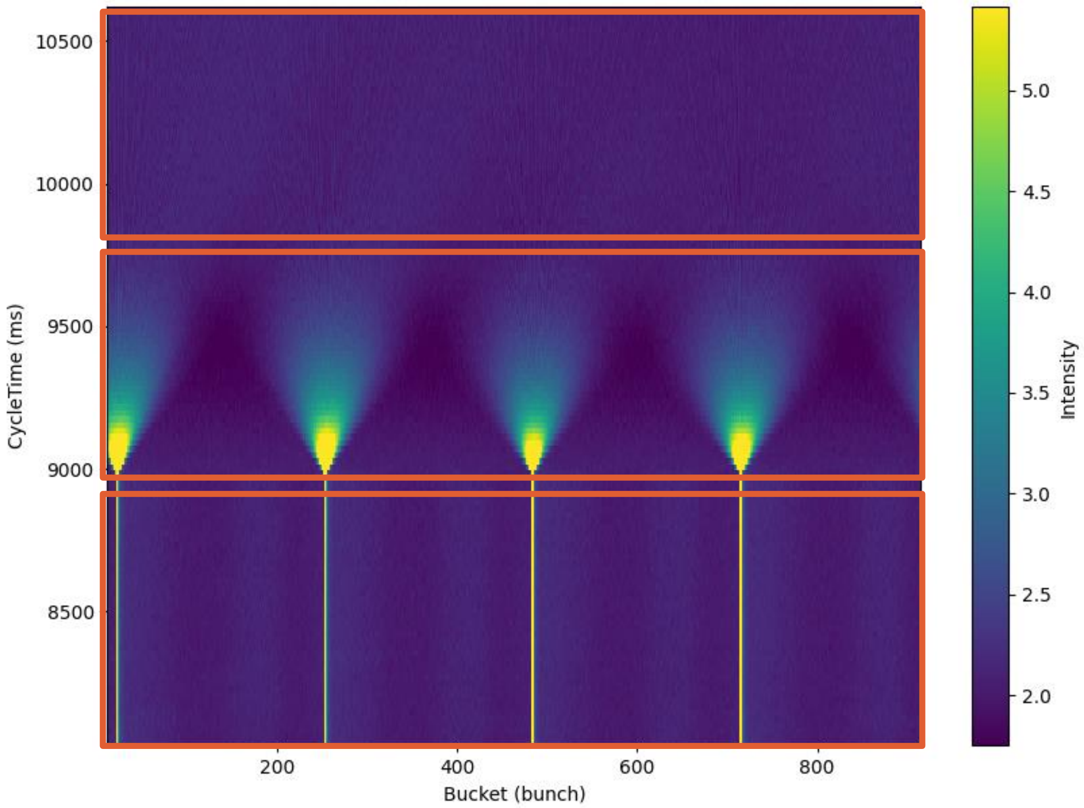
- **Loops are only active when enabled + timing**

	LHC protons	FT protons	AWAKE	LHC ions	FT ions
Phase loop	Yes	Yes	Yes	Yes	Yes
Synchro loop	Yes	Yes	Yes	Yes: injection + flat top	Yes
Radial loop	No	Yes: start ramp	No	Yes: start ramp	Yes: start ramp 2
Long. damper	High intensity / multi-batch	No	Yes: start ramp 3e11	No	No

RF manipulations for ions

RF manipulation for fixed target ions: De-bunching and recapture

- **4 ion bunches injected distributed along the ring**
  - 4x FM and AM modulation (Fixed Frequency Acceleration)
- **The RF is cut at intermediate flat top and the beam de-bunches**
  - We switch the Pickup source to high gain frontend (timing)
- **Recapture and acceleration to flat top with radial loop**



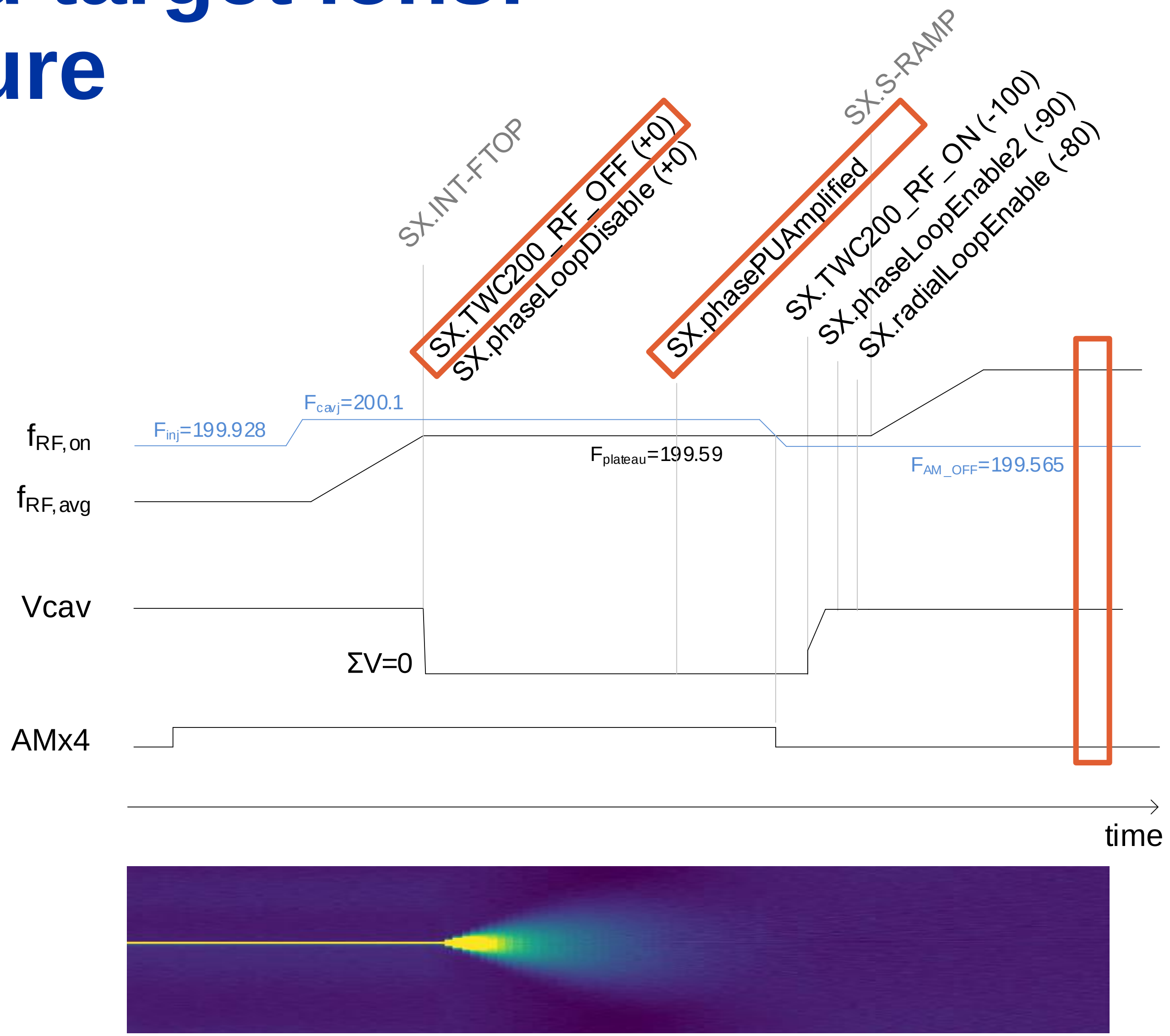
CCM -> BCTs

RF manipulation for fixed target ions: De-bunching and recapture

- **Cycle timing**

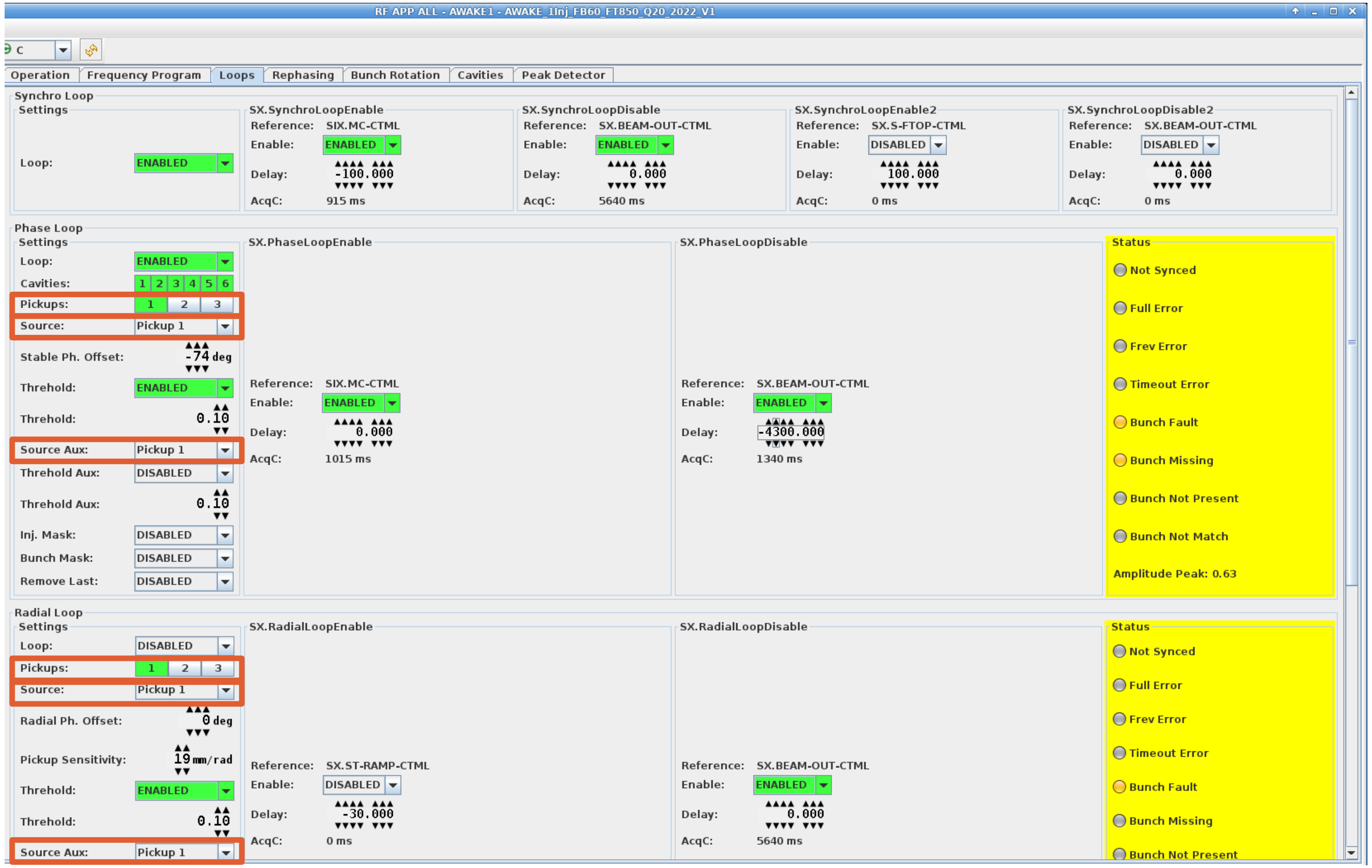
- **To disable de-bunching (for transverse setup)**
  - Disable TWC200 RF OFF
  - Disable Pickup switching

- **Bunch rotation is setup as for SFTPRO**



Beam loops: Pick-up selection

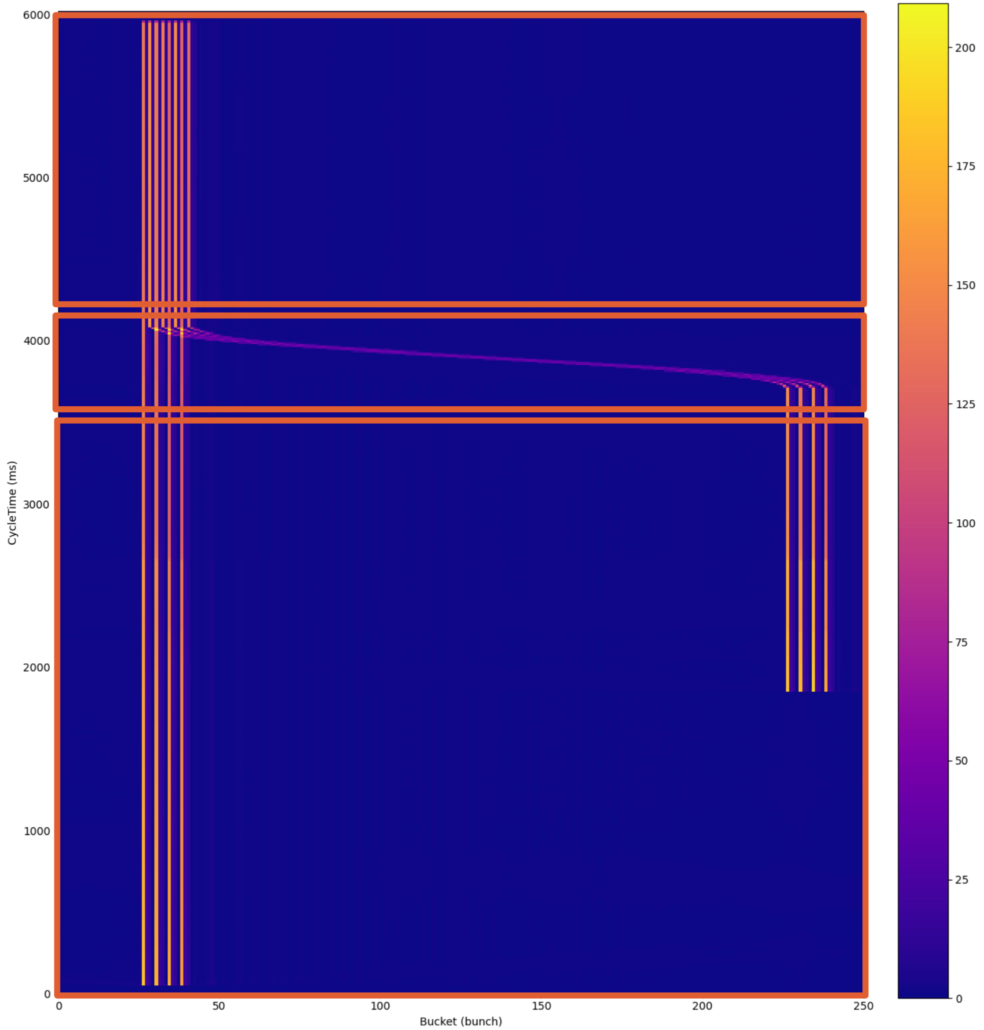
- **Enable pickups**
  - 1: standard
  - 2: High gain (iFT)
  - 3: Wideband
- **Select source and auxiliary source**
- **Setup timing**
  - SX.PhasePUAmplified
  - SX.PhasePUStandard
  - SX.RadialPUAmplified
  - SX.RadialPUStandard



CCM -> RF APP ALL: Loops tab

RF manipulation for LHC ions: slip stacking

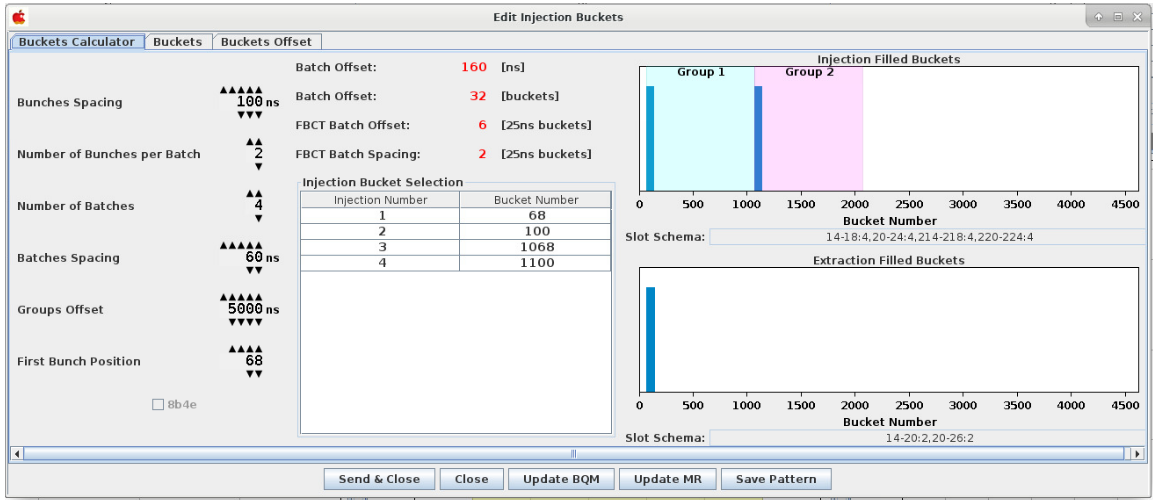
- **To decrease the bunch spacing from 100 ns to 50 ns**
- **Accelerate from 2 to 14 batches of ions with FFA (Fixed frequency Acceleration)**
- **Split the modulation around groups to control them separately (two phase/synchro loops)**
- **Apply frequency bumps, the LLRF detects the recapture time and uses the voltage jump settings**
- **Go back to synchro loop and rephasing**
- **Troubleshooting**
  - Setup filling pattern
  - Enable masks
  - Set pickup amplitude to half as usual (0.25 instead of 0.5)
  - How to enable/disable slip-stacking? (not that easy because of voltage functions)



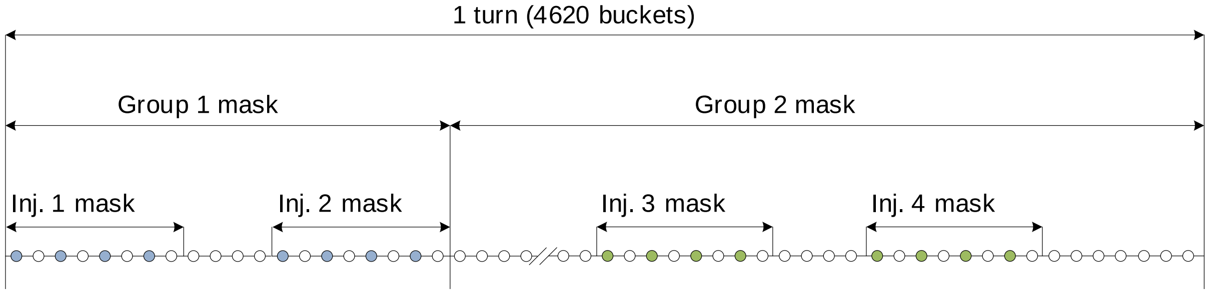
CCM -> BCTs

RF manipulation for LHC ions: slip stacking

- **Phase loop masks principle**
- **The phase loop error is an average over a range of bunches in the turn**
  - Two groups = two averages during slip stacking
  - When bunches from different groups start overlapping the reading is ignored



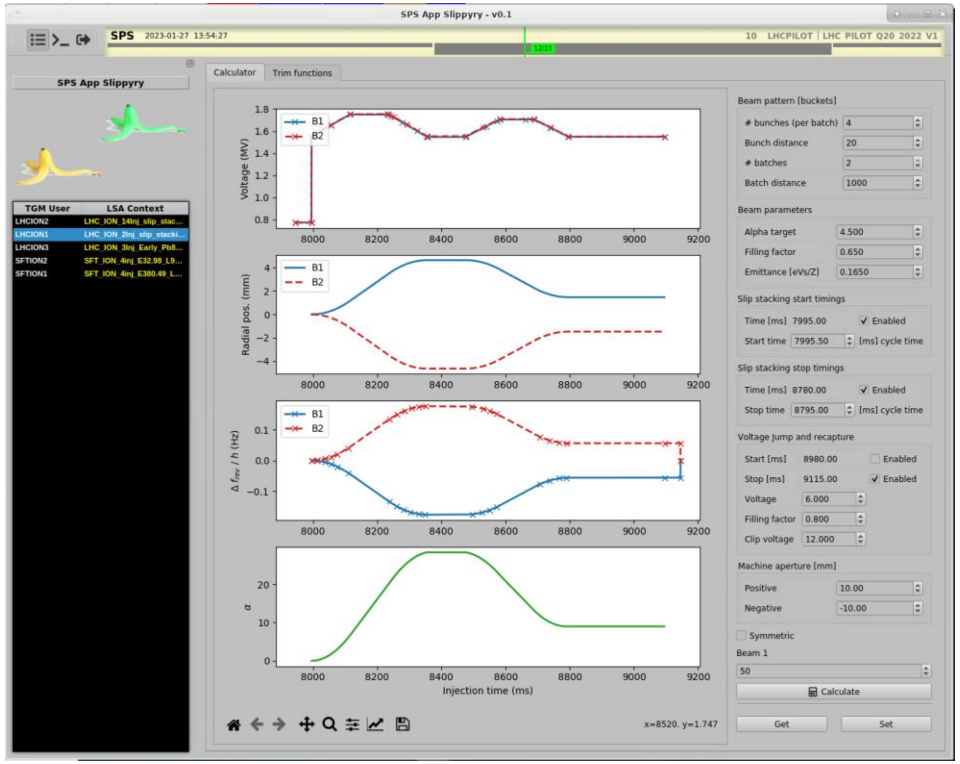
CCM -> RF APP ALL: operation tab -> edit injection bucket



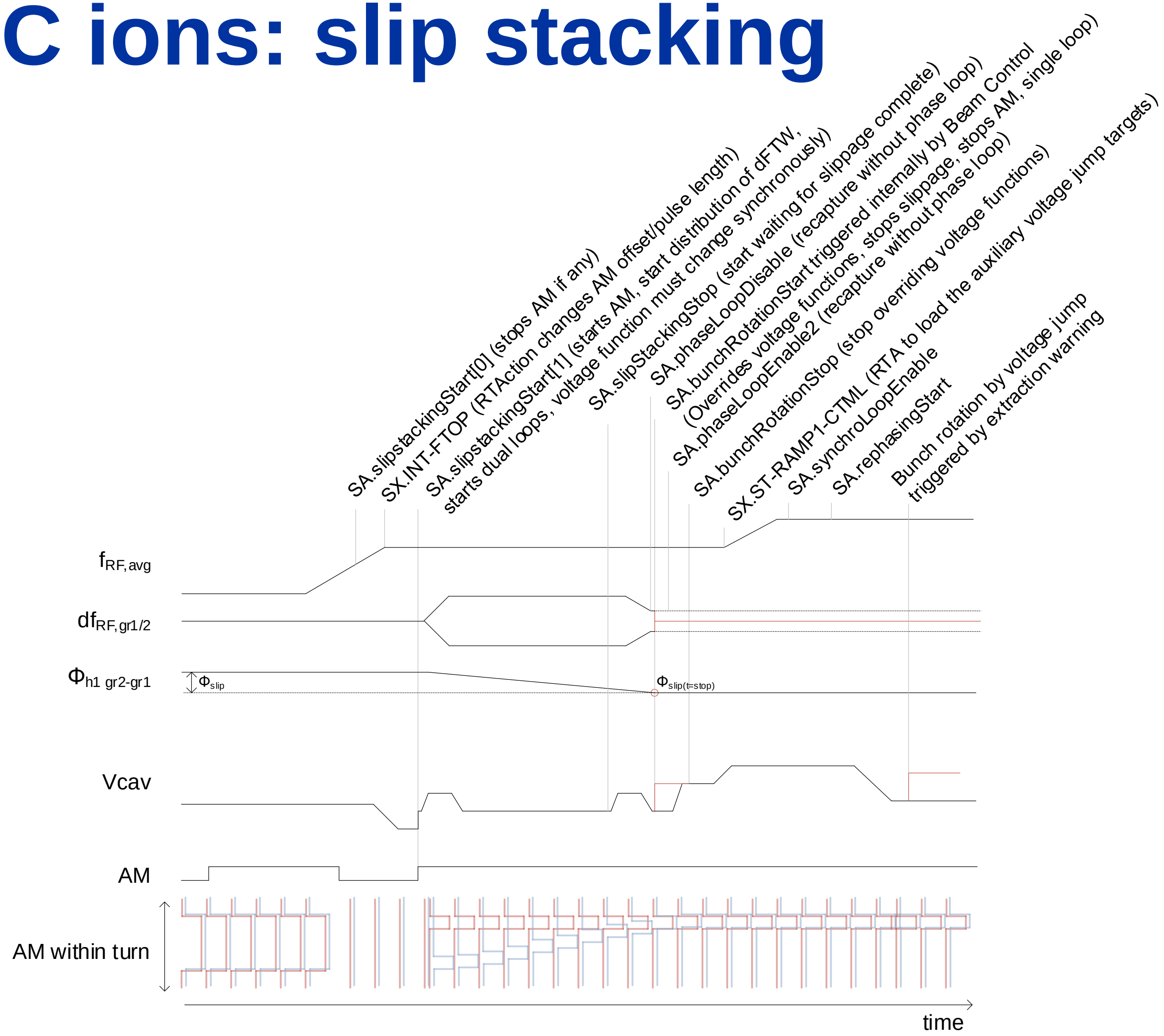
Phase loop masking during slip-stacking: the bunches in the red area are masked

RF manipulation for LHC ions: slip stacking

- **cycle timing**
- **Parameters generation**



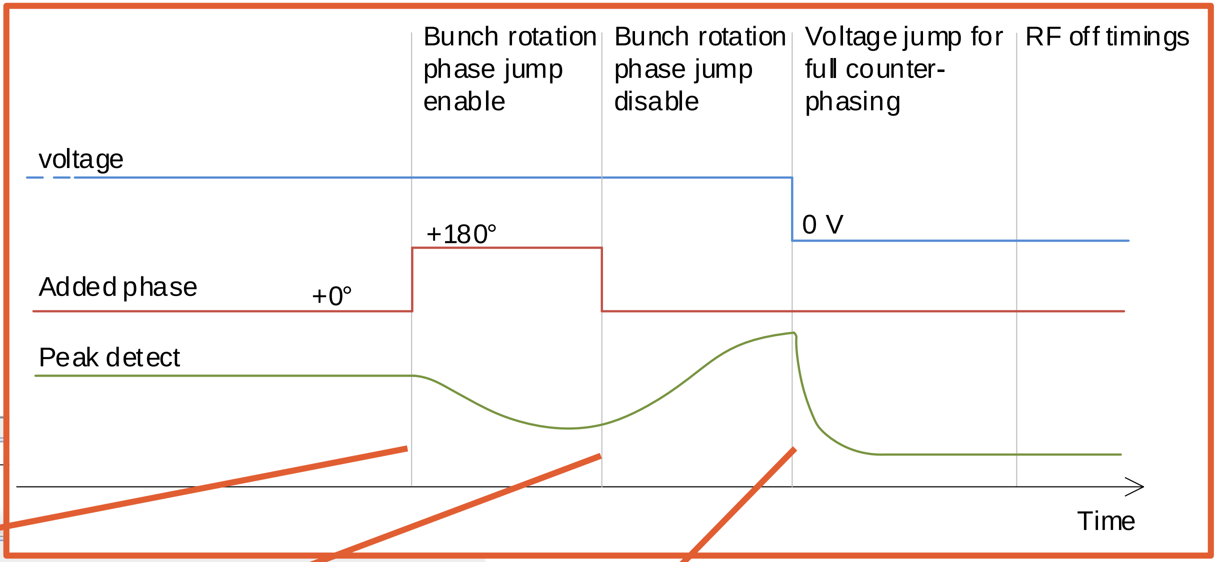
CCM -> slippyry: computes the frequency and voltage program for the slippage

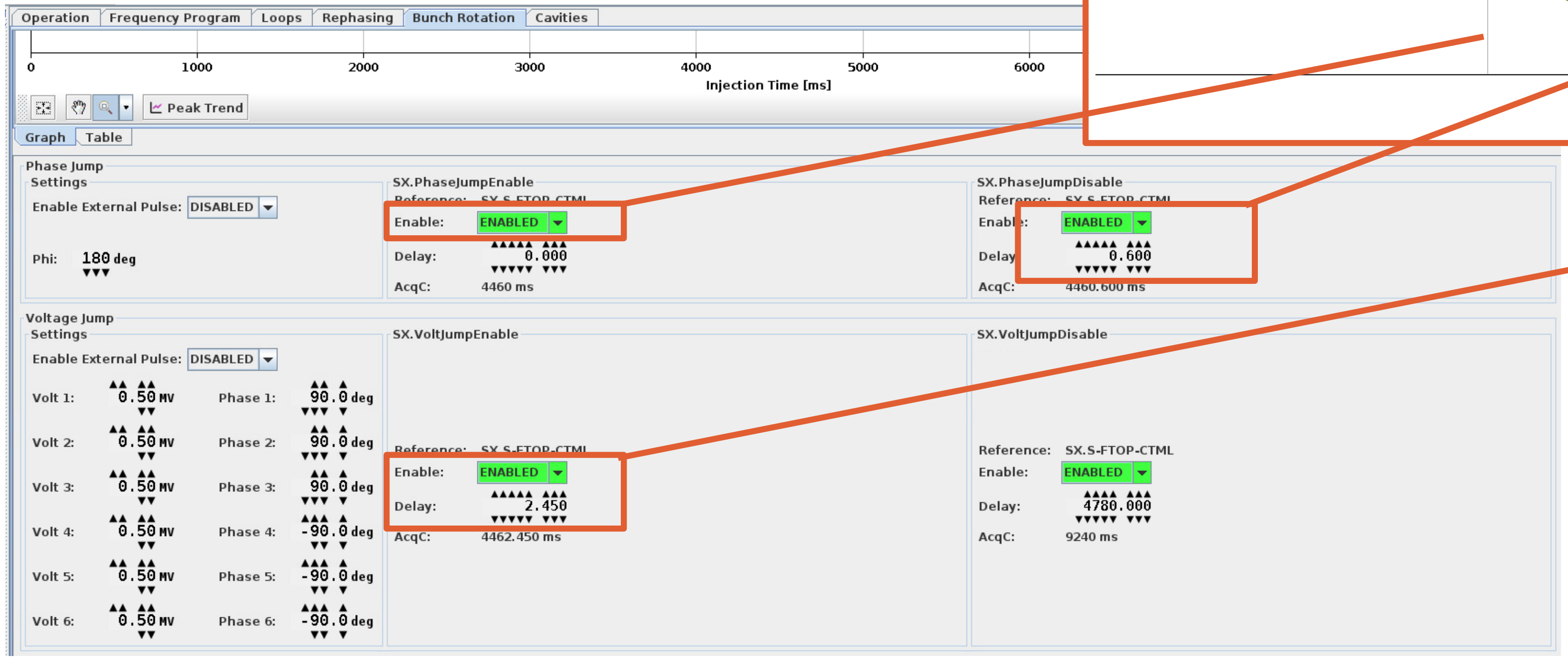


RF gymnastics

Bunch rotation for fixed target protons and ions

- **Jump to unstable phase**
- **To increase momentum spread before slow extraction (uniform spill)**

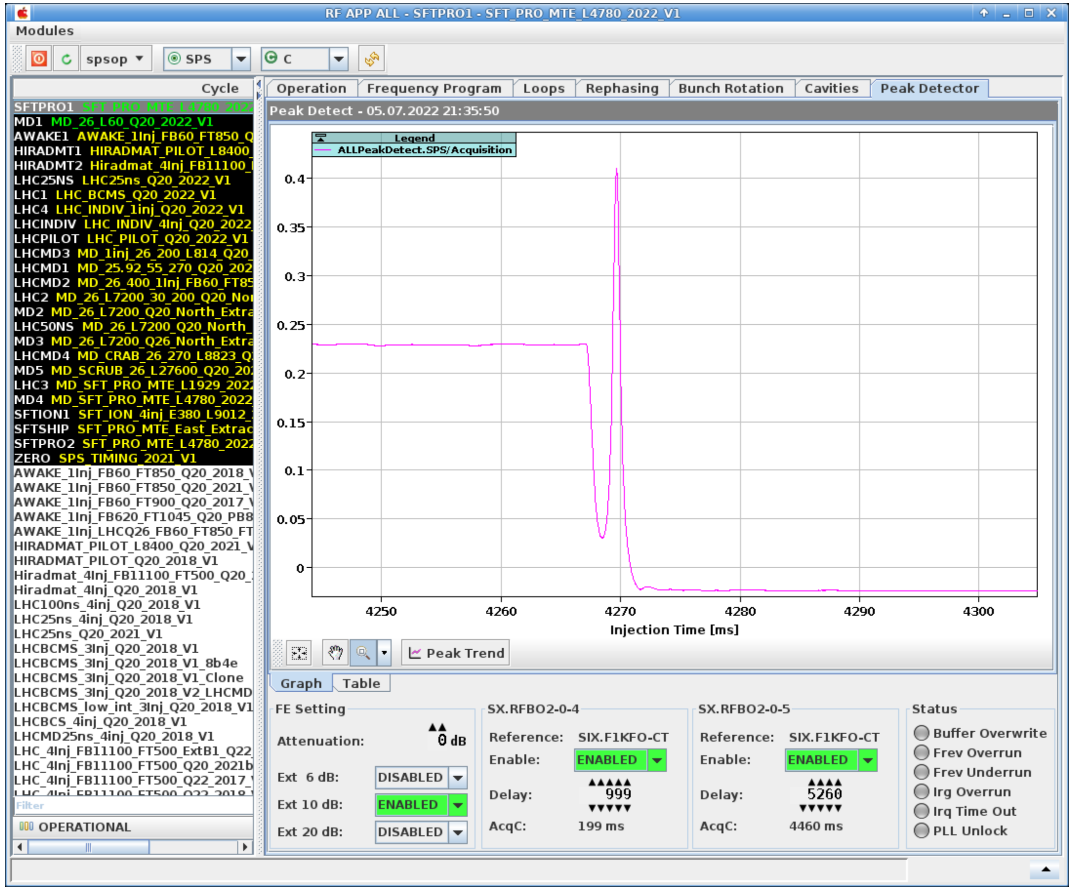




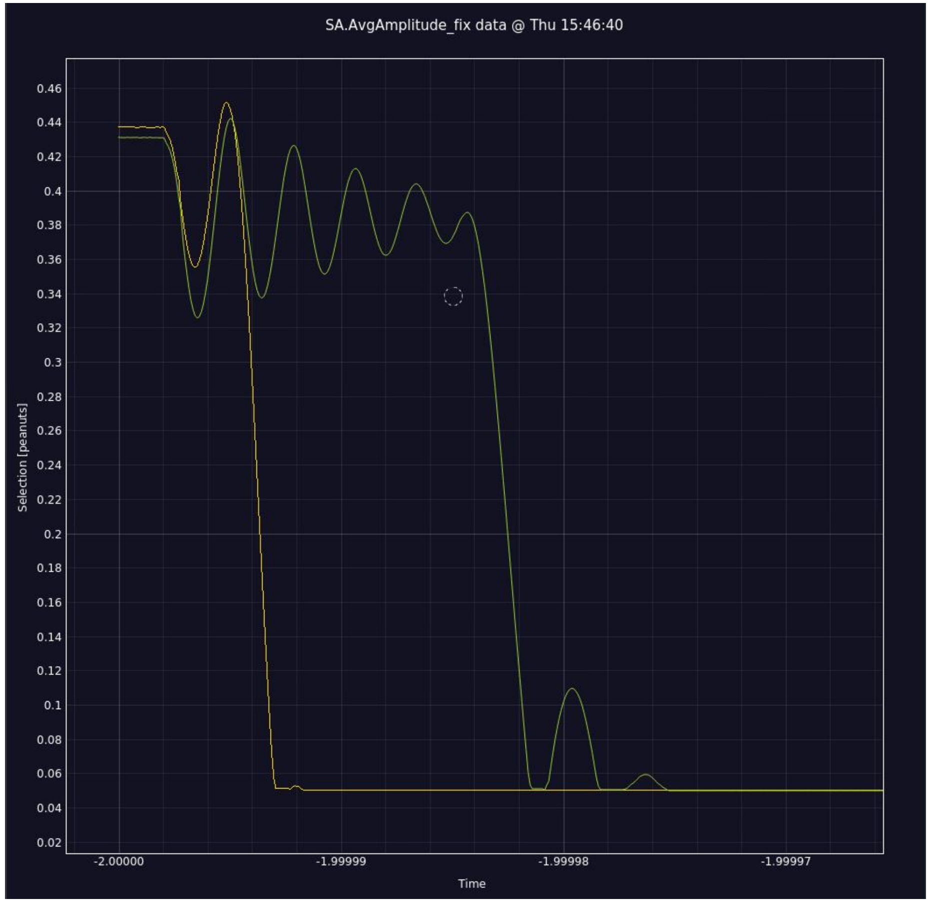
CCM -> RF APP ALL: Bunch rotation tab

- Ensure Voltages are set for full counter phasing (+/- 90°)
- Optimize the timings 2 and 3

Bunch rotation for fixed target protons and ions



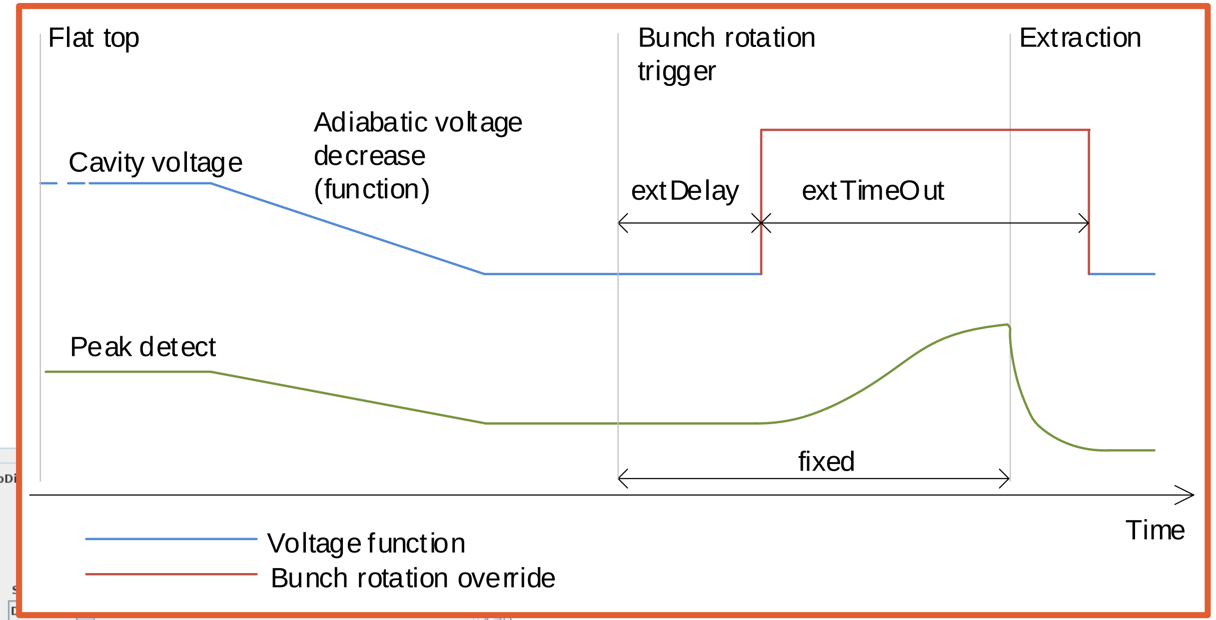
CCM -> RF APP ALL: Peak Detector tab

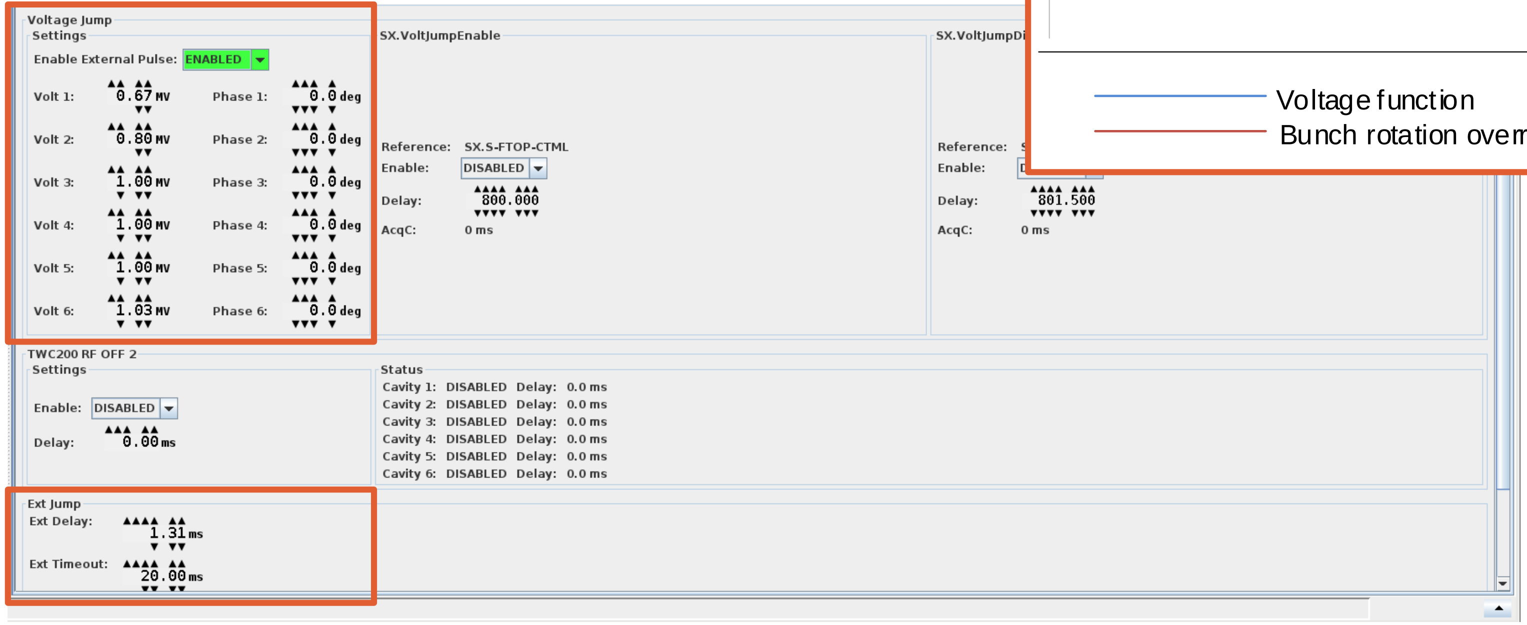


CCM -> Mega Drive: AvgAmplitude (useful when too little signal for peak detect)

Bunch rotation for AWAKE

- **By voltage jump**
- **To shorten the bunch at extraction**
- **The voltage/phase jump can be made less steep!**



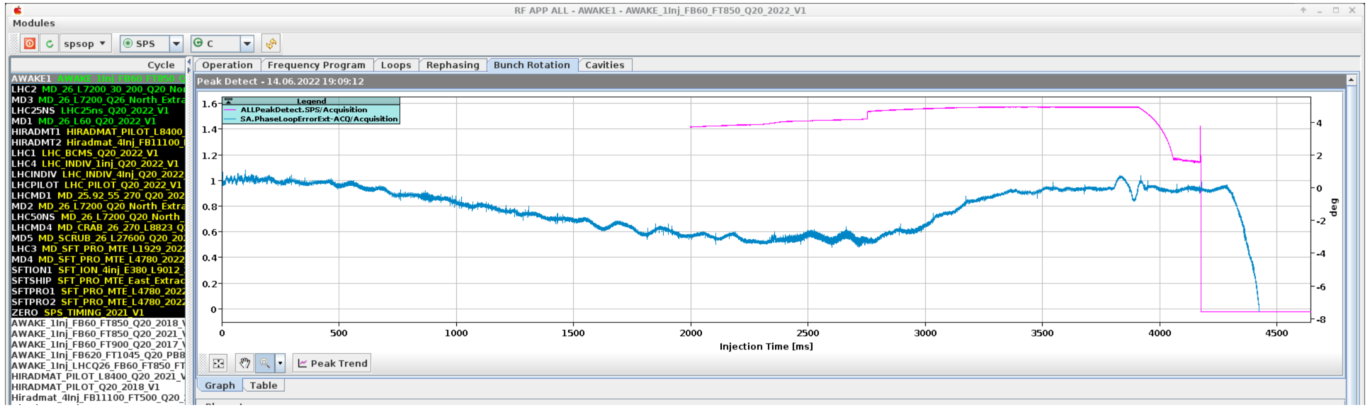


CCM -> RF APP ALL: Bunch rotation tab

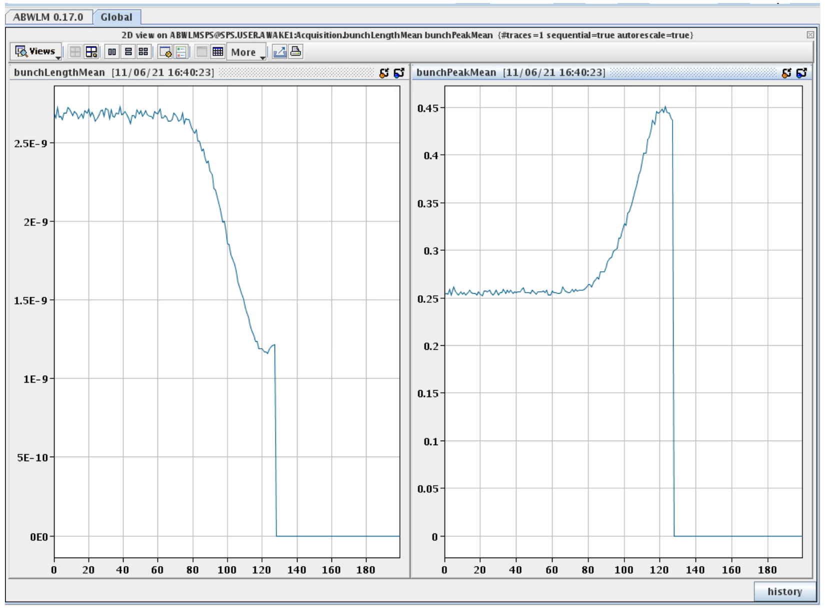
- Optimize the peak amplitude at extraction with Ext Delay
- Push voltages to a stable maximum
- The timeout avoids staying at maximum power

RF gymnastics: Bunch rotation for AWAKE

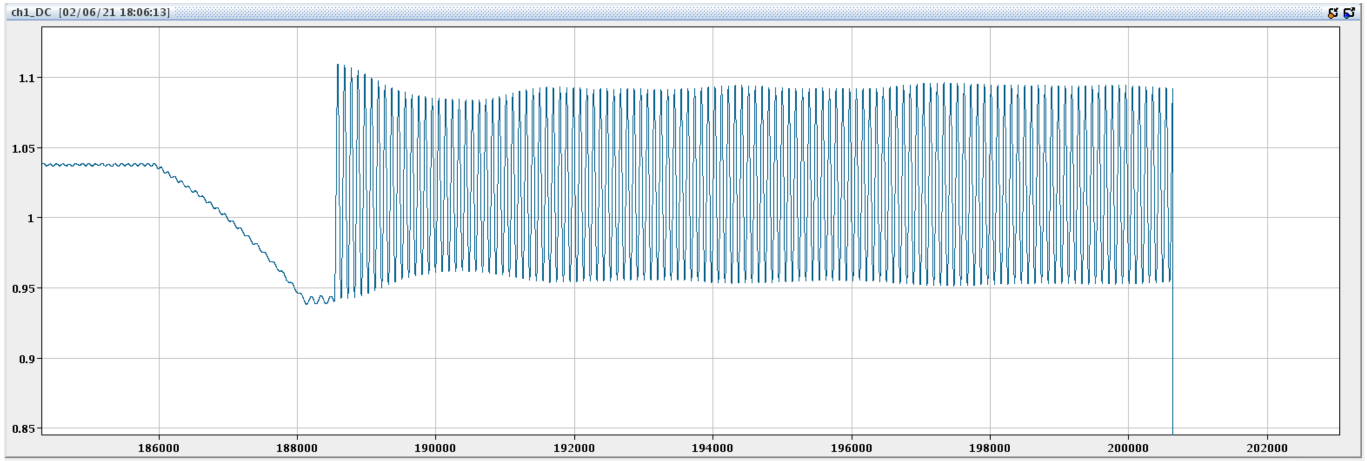
CCM -> RF APP ALL: bunch rotation tab (pink trace is bunch peak)



CCM -> ABWLM bunchLengthMean / bunchPeakMean



Peak detect: If the beam is not extracted, the bunch keeps oscillating (quadrupole)



References

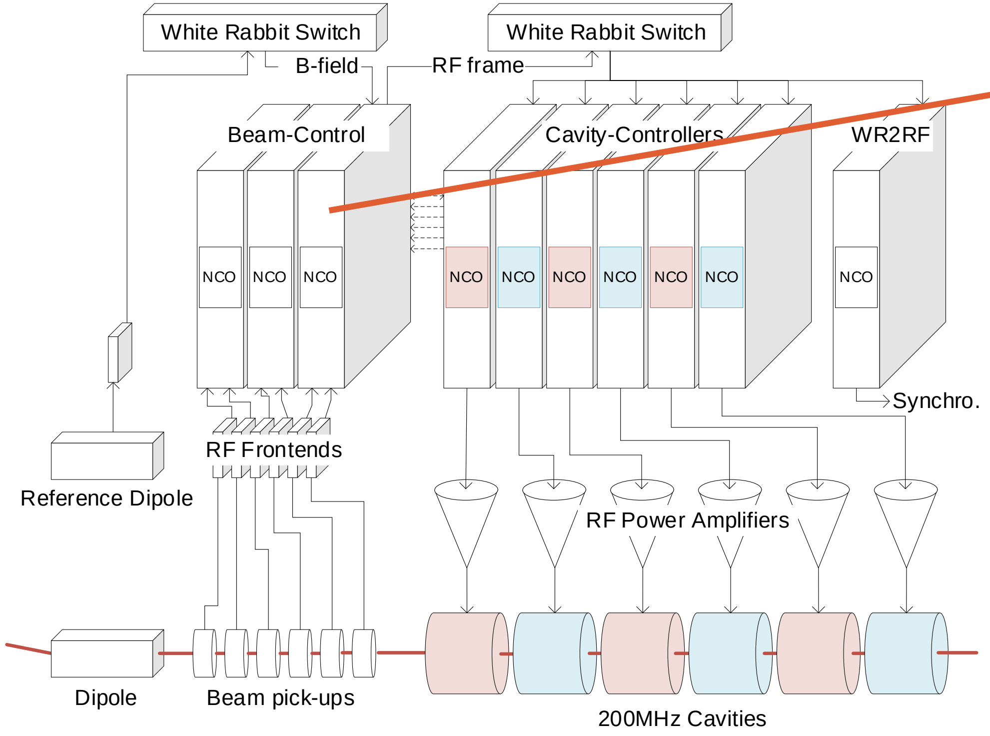
- **LLRF Piquet Training #1 and #2**
  - https://indico.cern.ch/event/1060631/ - LLRF and beams setting up
  - https://indico.cern.ch/event/1061720/ - Beam Control and the use of White Rabbit
- **Beam Control FW**
  - https://edms.cern.ch/document/2089558/0.1 - Detailed documentation on beam control LLRF
- **Details on pick-ups / Frontends and Firmware**
  - https://wikis.cern.ch/pages/viewpage.action?pageId=153716061
- **Details on WR network**
- **IPAC paper on the SPS beam control**
  - https://cds.cern.ch/record/2845762?ln=en - Beam Control
  - https://cds.cern.ch/record/2845800?ln=en - Lead Ions acceleration
- **Longitudinal dynamics and beam-based loops in synchrotrons**
  - https://cds.cern.ch/record/1415913?ln=en - CAS 2010 P. Baudrenghien
  - https://cds.cern.ch/record/702642?ln=en - CAS 2000 P. Baudrenghien

Summary of upgrades of 2023

- **Rephasing phase error around 0 even while using phase offset**
- **Rephasing optimizing only when beam present (issue with LHC ions)**
- **Loop freeze timing (EBC and more)**
- **Hopefully a cure for instabilities with synchro loop**
- **New Phase/Radial High Gain chains for Ions**
  - New pick-up switching mechanism
- **Robustness improvements**

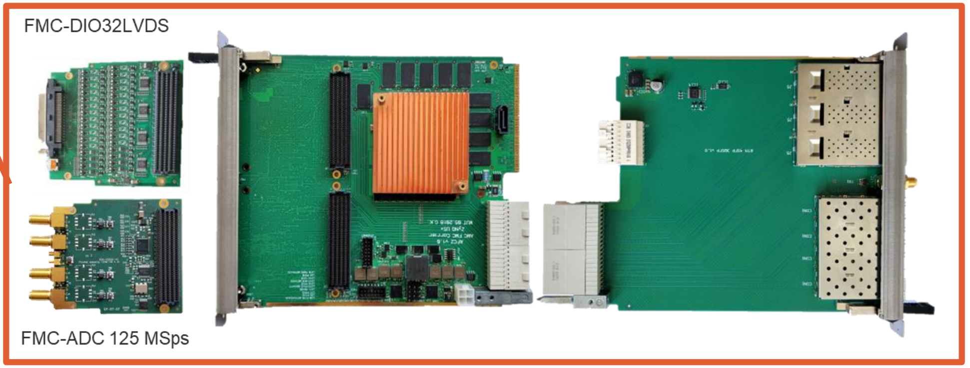
Beam Control Hardware

- **Where all these parameters end-up**



MicroTCA crate in BA3 Faraday Cage

FPGA Carrier board



Wideband longitudinal pickup coming soon

- **5 Giga Samples per second (instead of 125 MSps for narrowband)**
  - Bunch profile (Useful for observation too)
  - Bunch by bunch measurement for phase loop -> ideal for slip stacking

Wall current pickup (AEW)

Spare slides